Probability I: Fundamental of Probability Theory

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Preface

This is a reading note on measure-based probability written by the author with the support of ChatGPT. The materials in this note, presuming a basic knowledge of measure theory, is structured as follows: in chapter 1, I use the material from Bruce K. Driver's note *Probability Tools with Examples* to review measure theory, discuss its application in probability. I also use the material in this note to discuss the law of large numbers in chapter 2. The contents in chapter 3 about martingales are based on Achim Klenke's book *Probability Theory: A comprehensive course*.

Contents

1 Review of Measure Theory

1.1 Construction of measures

Definition 1.1 (Rings, semirings, and semialgebras). Let Ω be a set.

(i) A family $\mathcal{R} \subseteq \mathcal{P}(\Omega)$ is a *ring* if $\varnothing \in \mathcal{R}$ and $A \cup B, A \setminus B \in \mathcal{R}$ whenever $A, B \in \mathcal{R}$. It is an *algebra* if, in addition, $\Omega \in \mathcal{R}$.

(ii) A family $\mathcal{E} \subseteq \mathcal{P}(\Omega)$ is a *semiring* if $\varnothing \in \mathcal{E}$, it is closed under finite intersections, and every difference $A \setminus B$ with $A, B \in \mathcal{E}$ is a finite disjoint union of members of $\mathcal{E}$.

(iii) A semiring containing Ω is a *semialgebra*.

Theorem 1.2 (Ring generated by a semiring). If $\mathcal{E}$ is a semiring on Ω, then the ring generated by $\mathcal{E}$ is

$$\mathcal{R}(\mathcal{E}) = \left\{ \bigsqcup_{j=1}^{m} E_j : m \in \mathbb{N}_0,\ E_j \in \mathcal{E} \right\},$$

where the empty disjoint union is $\varnothing$. If $\mathcal{E}$ is a semialgebra, this ring contains Ω and is therefore the algebra generated by $\mathcal{E}$.

Proof. Let $\mathcal{R}_0$ denote the displayed family. It contains $\mathcal{E}$. Intersections of two members are finite disjoint unions of sets $E_i \cap F_j \in \mathcal{E}$. Repeatedly using the semiring difference property shows that

$$E_i \setminus \bigcup_j F_j$$

is a finite disjoint union of members of $\mathcal{E}$; taking the union over i proves closure under differences. Hence $\mathcal{R}_0$ is a ring. Every ring containing $\mathcal{E}$ contains all finite disjoint unions of its members, so $\mathcal{R}_0$ is the smallest such ring. The final assertion is immediate when $\Omega \in \mathcal{E}$. $\square$

Definition 1.3 (Premeasure). Let $\mathcal{R}$ be a ring on Ω. A map $\mu_0 : \mathcal{R} \to [0, \infty]$ is a *premeasure* if $\mu_0(\varnothing) = 0$ and, whenever (A_n) is a disjoint sequence in $\mathcal{R}$ whose union also belongs to $\mathcal{R}$,

$$\mu_0\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu_0(A_n).$$

The same definition is used on a semiring, with every set in the displayed identity required to belong to the semiring.

Theorem 1.4 (Extension from a semiring to its generated ring). Let $\mathcal{E}$ be a semiring and let $\chi : \mathcal{E} \to [0, \infty]$ be finitely additive in the following sense: if $E = \bigsqcup_{j=1}^{m} E_j$ with all sets in $\mathcal{E}$, then

$\chi(E) = \sum_{j=1}^m \chi(E_j)$. For $A = \bigsqcup_{j=1}^m E_j \in \mathcal{R}(\mathcal{E})$, define

$$\widetilde{\chi}(A) := \sum_{j=1}^{m} \chi(E_j).$$

Then $\widetilde{\chi}$ is well defined and finitely additive. If χ is a premeasure on $\mathcal{E}$, then $\widetilde{\chi}$ is a premeasure on $\mathcal{R}(\mathcal{E})$.

Proof. Suppose $A = \bigsqcup_{i=1}^m E_i = \bigsqcup_{j=1}^r F_j$. The sets $E_i \cap F_j$ form a common finite disjoint refinement. Finite additivity on $\mathcal{E}$ gives

$$\sum_i \chi(E_i) = \sum_{i,j} \chi(E_i \cap F_j) = \sum_j \chi(F_j),$$

so the definition is independent of the representation. Finite additivity of $\widetilde{\chi}$ follows by joining disjoint representations.

Now assume that χ is a premeasure. Let $A = \bigsqcup_{n \geq 1} A_n$ with $A, A_n \in \mathcal{R}(\mathcal{E})$. Refine A and every A_n into semiring pieces. Each semiring piece of A is a countable disjoint union of its intersections with the pieces of the A_n. Countable additivity of χ on $\mathcal{E}$, followed by Tonelli's theorem for nonnegative series, yields $\widetilde{\chi}(A) = \sum_n \widetilde{\chi}(A_n)$. □

Proposition 1.5 (Useful criterion for premeasures). Let μ_0 be finitely additive on a ring $\mathcal{R}$. Then μ_0 is a premeasure if and only if it is countably subadditive on $\mathcal{R}$: whenever $A \subseteq \bigcup_{n \geq 1} A_n$ with $A, A_n \in \mathcal{R}$,

$$\mu_0(A) \leq \sum_{n=1}^{\infty} \mu_0(A_n).$$

The same criterion holds on a semiring after passing to its generated ring by the preceding theorem.

Proof. A premeasure is countably subadditive after disjointifying the covering sets and intersecting them with A. Conversely, if $A = \bigsqcup_n A_n \in \mathcal{R}$, countable subadditivity gives $\mu_0(A) \leq \sum_n \mu_0(A_n)$. Finite additivity and monotonicity give, for each N, $\mu_0(A) \geq \sum_{n=1}^N \mu_0(A_n)$. Letting $N \to \infty$ gives the reverse inequality. □

Theorem 1.6 (Lebesgue–Stieltjes premeasure). Let $F : \mathbb{R} \to \mathbb{R}$ be nondecreasing. On the semiring

$$\mathcal{I} = \{\varnothing\} \cup \{(a, b] : a, b \in \mathbb{R}, a < b\},$$

define

$$\chi_F((a, b]) = F(b) - F(a), \qquad \chi_F(\varnothing) = 0.$$

Then χ_F is a premeasure if and only if F is right-continuous. In that case its Carathéodory extension is the Lebesgue–Stieltjes measure associated with F.

Proof. Finite additivity follows by telescoping after sorting endpoints. Assume first that F is right-continuous. By the preceding criterion, it is enough to show countable subadditivity. Suppose $(a,b] \subseteq \bigcup_n (a_n, b_n]$. Fix $\varepsilon > 0$. Choose $0 < \delta < b - a$ so that $F(a+\delta) - F(a) < \varepsilon$, and choose $\eta_n > 0$ so that $F(b_n + \eta_n) - F(b_n) < \varepsilon 2^{-n}$. The open intervals $(a_n, b_n + \eta_n)$ cover the compact interval $[a+\delta, b]$, so a finite subfamily covers it. The elementary finite interval-cover inequality for a nondecreasing function gives

$$F(b) - F(a+\delta) \leq \sum_{n \in J} \big(F(b_n + \eta_n) - F(a_n)\big)$$

for that finite index set J. Therefore

$$F(b) - F(a) \leq \sum_{n=1}^{\infty} \big(F(b_n) - F(a_n)\big) + 2\varepsilon.$$

Let $\varepsilon \downarrow 0$.

Conversely, suppose χ_F is a premeasure and let $h_n \downarrow 0$. The intervals $(x + h_{n+1}, x + h_n]$ are disjoint and their union is $(x, x+h_1]$. Countable additivity gives

$$F(x+h_1) - F(x) = \sum_{n=1}^{\infty} \big(F(x+h_n) - F(x+h_{n+1})\big).$$

The tail of this convergent nonnegative series is $F(x+h_n) - F(x)$, which therefore tends to zero. Thus F is right-continuous. □

Definition 1.7 (Outer measure). An *outer measure* on Ω is a map $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ such that

(i) $\mu^*(\varnothing) = 0$;

(ii) $A \subseteq B$ implies $\mu^*(A) \leq \mu^*(B)$;

(iii) $\mu^*(\bigcup_n A_n) \leq \sum_n \mu^*(A_n)$.

Theorem 1.8 (Outer measure generated by a covering cost). Let $\mathcal{E} \subseteq \mathcal{P}(\Omega)$ contain $\varnothing$, and let $\chi : \mathcal{E} \to [0, \infty]$ satisfy $\chi(\varnothing) = 0$. With the convention $\inf \varnothing = \infty$, define

$$\chi^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \chi(E_n) : A \subseteq \bigcup_{n=1}^{\infty} E_n, E_n \in \mathcal{E} \right\}.$$

Then χ^* is an outer measure.

Proof. The empty set is covered by the empty set, so $\chi^*(\varnothing) = 0$. Monotonicity is immediate because every cover of B also covers every $A \subseteq B$. For countable subadditivity, fix $\varepsilon > 0$ and, for each k, choose a cover $(E_{k,j})_j$ of A_k with total cost at most $\chi^*(A_k) + \varepsilon 2^{-k}$ whenever $\chi^*(A_k) < \infty$. The combined family covers $\bigcup_k A_k$. Taking the infimum and then letting $\varepsilon \downarrow 0$ proves the claim; cases with infinite right-hand side are automatic. □

Theorem 1.9 (Agreement with a premeasure). Let μ_0 be a premeasure on a ring $\mathcal{R}$, and let μ^* be the outer measure obtained by covering with sets in $\mathcal{R}$. Then

$$\mu^*(A) = \mu_0(A) \qquad (A \in \mathcal{R}).$$

Proof. The one-set cover gives $\mu^*(A) \le \mu_0(A)$. Conversely, if $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{R}$, disjointify the cover and intersect it with A. Countable additivity of μ_0 then gives $\mu_0(A) \le \sum_n \mu_0(A_n)$. Taking the infimum over all covers proves the reverse inequality. □

Definition 1.10 (Carathéodory measurability). For an outer measure μ^*, a set $E \subseteq \Omega$ is *μ^*-measurable* if

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \setminus E) \qquad \text{for every } A \subseteq \Omega.$$

Theorem 1.11 (Carathéodory's theorem). The family $\mathcal{M}_{\mu^*}$ of μ^*-measurable sets is a σ-algebra, and $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$ is a complete measure. If μ^* is generated by a premeasure μ_0 on a ring $\mathcal{R}$, then

$$\mathcal{R} \subseteq \mathcal{M}_{\mu^*} \quad \text{and} \quad \mu|_{\mathcal{R}} = \mu_0.$$

Proof. The reverse inequality in the defining equality follows from subadditivity, so only the forward inequality must be checked. The defining condition is stable under complements. Applying it successively shows that it is stable under finite unions. If (E_n) is disjoint and measurable, iteration gives

$$\mu^*(A) \ge \sum_{n=1}^{N} \mu^*(A \cap E_n) + \mu^*\left(A \setminus \bigcup_{n=1}^{N} E_n\right).$$

Let $N \to \infty$ and use subadditivity for the reverse inequality. Thus $\bigcup_n E_n$ is measurable, and arbitrary countable unions follow by disjointification. The same calculation proves countable additivity of the restriction.

If $N \in \mathcal{M}_{\mu^*}$ and $\mu(N) = 0$, then every $A \subseteq N$ has $\mu^*(A) = 0$. For every $B \subseteq \Omega$, subadditivity and monotonicity then give

$$\mu^*(B) \le \mu^*(B \cap A) + \mu^*(B \setminus A) = \mu^*(B \setminus A) \le \mu^*(B),$$

so A is measurable. Hence the measure is complete.

Finally, for $E \in \mathcal{R}$, cover an arbitrary set A by ring sets A_n. Finite additivity gives $\mu_0(A_n) = \mu_0(A_n \cap E) + \mu_0(A_n \setminus E)$. Taking infima proves the Carathéodory inequality for E. Agreement with μ_0 was proved above. □

Theorem 1.12 (Carathéodory extension and uniqueness). Every premeasure μ_0 on a ring $\mathcal{R}$ extends to a measure on $\sigma(\mathcal{R})$. If there are sets $E_k \in \mathcal{R}$ with $E_k \uparrow \Omega$ and $\mu_0(E_k) < \infty$, then this extension is unique on $\sigma(\mathcal{R})$.

Proof. Existence follows by restricting the induced outer measure to the Carathéodory measurable sets. For uniqueness, let μ and ν be two extensions and fix k. The class

$$\mathcal{D}_k = \{A \in \sigma(\mathcal{R}) : \mu(A \cap E_k) = \nu(A \cap E_k)\}$$

is a Dynkin system containing the π-system $\mathcal{R}$, because both restricted measures have finite total mass on E_k. Hence $\mathcal{D}_k = \sigma(\mathcal{R})$. Letting $k \to \infty$ and using continuity from below gives $\mu(A) = \nu(A)$ for every $A \in \sigma(\mathcal{R})$. □

1.2 Radon measures

Proposition 1.13 (Continuity of a measure). Let μ be a measure.

(i) If $A_n \uparrow A$, then $\mu(A_n) \uparrow \mu(A)$.

(ii) If $A_n \downarrow A$ and $\mu(A_1) < \infty$, then $\mu(A_n) \downarrow \mu(A)$.

Conversely, a finitely additive set function on an algebra is a premeasure if it is continuous from below. If its total mass is finite, this is equivalent to continuity from above, and also to continuity at the empty set.

Proof. For the first assertion, write A as the disjoint union of A_1 and the increments $A_n \setminus A_{n-1}$. The second follows by applying the first to $A_1 \setminus A_n$. The converses follow by applying the corresponding continuity property to partial unions of a disjoint sequence; under finite total mass, complements convert increasing sequences into decreasing ones. □

Definition 1.14 (Radon measure). A Borel measure μ on a locally compact Hausdorff space X is a *Radon measure* if it is finite on compact sets and inner regular:

$$\mu(B) = \sup\{\mu(K) : K \subseteq B,\ K \text{ compact}\} \qquad (B \in \mathcal{B}(X)).$$

On metric spaces this is equivalent, for locally finite Borel measures, to being both inner regular by compact sets and outer regular by open sets.

Theorem 1.15 (Regularity on Euclidean space). Every locally finite Borel measure on $\mathbb{R}^d$ is Radon. In particular, every finite Borel measure on $\mathbb{R}^d$ is inner regular and outer regular.

Proof. First assume that μ is finite. Let $\mathcal{R}$ be the class of Borel sets B such that for every $\varepsilon > 0$ there are a compact set $K \subseteq B$ and an open set $G \supseteq B$ with $\mu(G \setminus K) < \varepsilon$. Closed sets belong to $\mathcal{R}$: use the compact exhaustion $F \cap \overline{B}(0, n)$ for inner approximation and the open neighbourhoods $\{x : d(x, F) < 1/n\}$ for outer approximation. The class $\mathcal{R}$ is a σ-algebra: complements interchange inner and outer approximations, while countable unions are handled by approximating a finite initial union and making the remaining total measure small. Since closed sets generate the Borel σ-algebra, every Borel set belongs to $\mathcal{R}$.

For a locally finite measure, apply the finite result to the restrictions of μ to increasing compact balls and then let the radius tend to infinity. □

Proposition 1.16 (Distribution function of a locally finite measure). Let μ be a locally finite Borel measure on $\mathbb{R}$. Define

$$F_\mu(x) = \begin{cases} \mu((0,x]), & x \geq 0, \\ -\mu((x,0]), & x < 0. \end{cases}$$

Then F_μ is nondecreasing and right-continuous, and

$$\mu((a,b]) = F_\mu(b) - F_\mu(a) \qquad (a < b).$$

Conversely, every nondecreasing right-continuous function determines a unique locally finite Lebesgue–Stieltjes measure through this identity.

Proof. The increment identity follows by splitting intervals at zero and using finite additivity. Monotonicity is immediate. If $x_n \downarrow x$, then $(x, x_n] \downarrow \varnothing$ and local finiteness gives $F_\mu(x_n) - F_\mu(x) \to 0$. The converse is the Lebesgue–Stieltjes construction followed by uniqueness of the σ-finite extension. □

Theorem 1.17 (Uniqueness of Lebesgue measure). If μ is a translation-invariant, locally finite Borel measure on $\mathbb{R}$, then

$$\mu = c\lambda, \qquad c = \mu([0,1)),$$

where λ is Lebesgue measure. In particular, the normalization $\mu([0,1)) = 1$ determines Lebesgue measure uniquely.

Proof. Translation invariance and finite additivity give $\mu([0,n)) = nc$ and $\mu([0,1/n)) = c/n$. Hence $\mu([0,q)) = cq$ for every nonnegative rational q. Continuity from below and above extends this identity to every real $q \geq 0$. Translation invariance then gives $\mu([a,b)) = c(b-a)$ for all $a < b$. The half-open intervals form a π-system generating $\mathcal{B}(\mathbb{R})$, and uniqueness of measures on that generating class proves $\mu = c\lambda$. □

1.3 Random variables and measurability

Definition 1.18 (Pullback and pushforward). Let $f : \Omega \to S$ be a map.

(i) For a family $\mathcal{C} \subseteq \mathcal{P}(S)$, its pullback is $f^{-1}(\mathcal{C}) = \{f^{-1}(C) : C \in \mathcal{C}\}$.

(ii) If $(\Omega, \mathcal{F}, P)$ is a probability space and $f : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ is measurable, the pushforward law is

$$f_\# P(B) = P(f \in B) = P(f^{-1}(B)), \qquad B \in \mathcal{S}.$$

Proposition 1.19 (Generated pullbacks). For every family $\mathcal{C} \subseteq \mathcal{P}(S)$,

$$f^{-1}(\sigma(\mathcal{C})) = \sigma(f^{-1}(\mathcal{C})).$$

Consequently, to prove that $f : (\Omega, \mathcal{F}) \to (S, \sigma(\mathcal{C}))$ is measurable, it is enough to check $f^{-1}(C) \in \mathcal{F}$ for $C \in \mathcal{C}$.

Proof. The class of sets $B \subseteq S$ whose inverse image belongs to $\sigma(f^{-1}(\mathcal{C}))$ is a σ-algebra containing $\mathcal{C}$. This gives one inclusion; the other follows because inverse images preserve complements and countable unions. □

Theorem 1.20 (Doob–Dynkin factorization). Let $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ be measurable and let $Y : \Omega \to \mathbb{R}$ be $\sigma(X)$-measurable. Then there is an $\mathcal{S}/\mathcal{B}(\mathbb{R})$-measurable function $g : S \to \mathbb{R}$ such that

$$Y = g \circ X.$$

The same conclusion holds for maps into any standard Borel space.

Proof. For each rational q, choose $A_q \in \mathcal{S}$ such that $\{Y < q\} = X^{-1}(A_q)$. Replacing A_q by $\bigcup_{r<q,\, r\in\mathbb{Q}} A_r$, we may assume that the family is increasing and that its inverse images are unchanged. Define the extended-real measurable function

$$\tilde{g}(s) = \inf\{q \in \mathbb{Q} : s \in A_q\}, \qquad \inf\varnothing = +\infty.$$

For rational q, $\{\tilde{g} < q\} = \bigcup_{r<q} A_r$, so $\tilde{g}$ is measurable. Put $g = \tilde{g}$ on $\{\tilde{g} \in \mathbb{R}\}$ and $g = 0$ elsewhere. Then $g : S \to \mathbb{R}$ is measurable, and the rational sublevel sets show that $g(X) = Y$. The standard Borel version follows by transporting the argument through a Borel isomorphism with a Borel subset of $[0, 1]$. □

Theorem 1.21 (Integration with respect to a law). Let $X : (\Omega, \mathcal{F}, P) \to (S, \mathcal{S})$ be measurable and let $\mu_X = X_\# P$. For every nonnegative measurable h, and for every integrable measurable h,

$$\mathbb{E}[h(X)] = \int_S h(x)\, \mu_X(\mathrm{d}x).$$

Moreover, $h(X)$ is integrable if and only if $\int_S |h|\, \mathrm{d}\mu_X < \infty$.

Proof. The identity is immediate for indicators, extends to nonnegative simple functions by linearity, to nonnegative functions by monotone convergence, and to integrable functions by positive and negative parts. □

Lemma 1.22 (First Borel–Cantelli lemma). For measurable sets (A_n) in a measure space,

$$\sum_{n=1}^{\infty} \mu(A_n) < \infty \quad \Longrightarrow \quad \mu(\limsup_{n\to\infty} A_n) = 0,$$

where $\limsup_n A_n = \bigcap_{m\geq 1} \bigcup_{n\geq m} A_n$ is the event that infinitely many A_n occur.

Proof. For every m,

$$\mu\left(\bigcup_{n\geq m} A_n\right) \leq \sum_{n\geq m} \mu(A_n).$$

The right-hand side tends to zero, and continuity from above gives the result. $\square$

Theorem 1.23 (Differentiation under the integral sign). Let $I \subseteq \mathbb{R}$ be open, let $(S, \mathcal{S}, \mu)$ be a measure space, and let $f : I \times S \to \mathbb{R}$ satisfy:

(i) for every $x \in I$, $f(x, \cdot)$ is measurable and integrable;

(ii) for almost every s, the map $x \mapsto f(x, s)$ is differentiable;

(iii) there is $h \in L^1(\mu)$ such that $|\partial_x f(x, s)| \leq h(s)$ for all $x \in I$ and almost every s.

Then $F(x) = \int f(x, s)\, \mu(\mathrm{d}s)$ is differentiable and

$$F'(x) = \int \partial_x f(x, s)\, \mu(\mathrm{d}s).$$

Proof. Fix $x \in I$ and let $t_n \to 0$ with $x + t_n \in I$. For almost every s, the difference quotient converges to $\partial_x f(x, s)$. The mean-value theorem bounds its absolute value by $h(s)$. Dominated convergence therefore gives the derivative along every such sequence, and hence the asserted derivative. $\square$

Definition 1.24 (Multiplicative class). A family $\mathcal{M}$ of bounded real-valued functions on S is *multiplicative* if $fg \in \mathcal{M}$ whenever $f, g \in \mathcal{M}$. Let $\mathcal{H}(\mathcal{M})$ be the smallest vector space of bounded functions that contains 1 and $\mathcal{M}$ and is closed under bounded pointwise convergence.

Theorem 1.25 (Functional monotone-class theorem). If $\mathcal{M}$ is multiplicative, then $\mathcal{H}(\mathcal{M})$ contains every bounded $\sigma(\mathcal{M})$-measurable function, where $\sigma(\mathcal{M})$ is the smallest σ-algebra making every member of $\mathcal{M}$ measurable.

Proof. For fixed $f \in \mathcal{H}(\mathcal{M})$, let $\mathcal{H}_f = \{g : fg \in \mathcal{H}(\mathcal{M})\}$. This class is a vector space closed under bounded convergence. First take $f \in \mathcal{M}$; the multiplicative property shows that $\mathcal{M} \subseteq \mathcal{H}_f$, hence $\mathcal{H}(\mathcal{M}) \subseteq \mathcal{H}_f$. Reversing the roles of f and g shows that $\mathcal{H}(\mathcal{M})$ is itself closed under products. Now let $\mathcal{D} = \{A \subseteq S : \mathbf{1}_A \in \mathcal{H}(\mathcal{M})\}$. The class $\mathcal{D}$ is a Dynkin system. For $f \in \mathcal{M}$, polynomial functions of f belong to $\mathcal{H}(\mathcal{M})$ by product closure. Uniform polynomial approximation on the compact range of f therefore puts every bounded continuous function of f in $\mathcal{H}(\mathcal{M})$. Approximating the indicator of $(-\infty, a]$ by bounded continuous cutoff functions shows that

$$\{f \leq a\} \in \mathcal{D} \qquad (f \in \mathcal{M},\ a \in \mathbb{Q}).$$

Because $\mathcal{H}(\mathcal{M})$ is closed under products, $\mathcal{D}$ is also closed under finite intersections. It therefore contains the π-system of finite intersections of these sublevel sets, which generates $\sigma(\mathcal{M})$. The

π–λ theorem gives $\sigma(\mathcal{M}) \subseteq \mathcal{D}$. Finally, bounded measurable functions are bounded pointwise limits of simple functions, completing the proof. $\square$

Proposition 1.26. The Borel σ-algebra on $\mathbb{R}^d$ is generated by $C_c(\mathbb{R}^d)$:

$$\sigma(C_c(\mathbb{R}^d)) = \mathcal{B}(\mathbb{R}^d).$$

Proof. Every continuous function is Borel measurable, so one inclusion is clear. For the converse, the claim is immediate for $U = \mathbb{R}^d$. For any other open set U, choose continuous cutoff functions η_m that are positive on $B(0, m)$ and vanish outside $B(0, m+1)$, and put

$$f_m(x) = \eta_m(x) \min\{1, d(x, U^c)\}.$$

Then $f_m \in C_c(\mathbb{R}^d)$ and $U = \bigcup_m \{f_m > 0\}$. Thus every open set belongs to the generated σ-algebra. $\square$

Theorem 1.27 (Determination by compactly supported test functions)**.** Let μ and ν be finite Borel measures on $\mathbb{R}^d$. If

$$\int f \, \mathrm{d}\mu = \int f \, \mathrm{d}\nu \qquad \text{for every } f \in C_c(\mathbb{R}^d),$$

and $\mu(\mathbb{R}^d) = \nu(\mathbb{R}^d)$, then $\mu = \nu$. In particular, the extra mass condition is automatic for probability measures.

Proof. Let $\mathcal{M} = C_c(\mathbb{R}^d) \cup \{1\}$; this is multiplicative. The class of bounded Borel functions f for which the two integrals agree is a vector space containing $\mathcal{M}$ and closed under bounded pointwise convergence. The functional monotone-class theorem and the preceding proposition imply that it contains every bounded Borel function. Apply this to indicators. $\square$

1.4 Product measures

Definition 1.28 (Product σ-algebra)**.** For measurable spaces $(S_i, \mathcal{S}_i)$, $i \in I$, the product σ-algebra on $\prod_{i \in I} S_i$ is

$$\bigotimes_{i \in I} \mathcal{S}_i := \sigma\{\pi_i^{-1}(A) : i \in I,\ A \in \mathcal{S}_i\},$$

where π_i is the ith coordinate projection.

Proposition 1.29 (Coordinate criterion)**.** A map $X : (\Omega, \mathcal{F}) \to (\prod_{i \in I} S_i, \bigotimes_{i \in I} \mathcal{S}_i)$ is measurable if and only if every coordinate $\pi_i \circ X$ is measurable.

Proof. Necessity follows because each projection is measurable. Conversely, the inverse image under X of every generating cylinder $\pi_i^{-1}(A)$ is $(\pi_i \circ X)^{-1}(A) \in \mathcal{F}$. $\square$

Theorem 1.30 (Product measure and Tonelli–Fubini). Let $(S, \mathcal{S}, \mu)$ and $(T, \mathcal{T}, \nu)$ be σ-finite measure spaces. There is a unique measure $\mu \otimes \nu$ on $\mathcal{S} \otimes \mathcal{T}$ satisfying

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B).$$

If $f : S \times T \to [0, \infty]$ is measurable, then its sections are measurable and

$$\int f \, \mathrm{d}(\mu \otimes \nu) = \int_S \left(\int_T f(s,t) \, \nu(\mathrm{d}t) \right) \mu(\mathrm{d}s) = \int_T \left(\int_S f(s,t) \, \mu(\mathrm{d}s) \right) \nu(\mathrm{d}t).$$

If f is integrable, the same identities hold for f, both iterated integrals are absolutely integrable, and the order of integration may be interchanged.

Proof. Construct the product measure from the rectangle premeasure. The identities hold first for rectangle indicators, then for nonnegative simple functions, and then for nonnegative measurable functions by monotone convergence. For integrable f, apply the nonnegative result to f^+ and f^-, or first to $|f|$, to justify all subtractions and both iterated integrals. □

1.5 Kolmogorov's extension theorem

Theorem 1.31 (Kolmogorov extension on the compact cube). Let $Q = [0,1]^{\mathbb{N}}$. For each n, let μ_n be a probability measure on $[0,1]^n$. Assume projective consistency:

$$\mu_{n+1}(B \times [0,1]) = \mu_n(B) \qquad (B \in \mathcal{B}([0,1]^n)).$$

Then there is a unique probability measure μ on $(Q, \bigotimes_{n \geq 1} \mathcal{B}([0,1]))$ such that

$$\mu(\pi_{1:n}^{-1}(B)) = \mu_n(B) \qquad (n \geq 1).$$

Proof. Define μ_0 on the cylinder algebra by $\mu_0(\pi_{1:n}^{-1}(B)) = \mu_n(B)$. Consistency makes this well defined and finitely additive. It remains to prove continuity at the empty set. Suppose C_k is a decreasing sequence of cylinders with $\inf_k \mu_0(C_k) > 0$. By regularity of the finite-dimensional measures, choose compact cylinders $K_k \subseteq C_k$ so that $\mu_0(C_k \setminus K_k) < 2^{-k-1} \inf_j \mu_0(C_j)$. Since $C_m \subseteq C_k$ for $k \leq m$,

$$\begin{aligned} \mu_0\left(C_m \setminus \bigcap_{k=1}^m K_k \right) &\leq \sum_{k=1}^m \mu_0(C_m \setminus K_k) \\ &\leq \sum_{k=1}^m \mu_0(C_k \setminus K_k) < \frac{1}{2} \inf_j \mu_0(C_j). \end{aligned}$$

Thus every finite intersection $\bigcap_{k=1}^m K_k$ has positive measure and is nonempty. These intersections are compact and decreasing in the compact space Q, so their full intersection is nonempty. Hence $\bigcap_k C_k \neq \varnothing$. The contrapositive is continuity at the empty set, so μ_0 is a premeasure. Carathéodory's theorem gives existence, and uniqueness follows because cylinders form a generating π-system. $\square$

Definition 1.32 (Standard Borel space). A measurable space $(S, \mathcal{S})$ is *standard Borel* if it is measurably isomorphic to a Borel subset of a Polish space. Equivalently, it is measurably isomorphic to a Borel subset of $[0, 1]$.

Theorem 1.33 (Kolmogorov extension for countable standard Borel products). Let $(S_n, \mathcal{S}_n)$ be standard Borel spaces. Suppose μ_n is a probability measure on $(S_1 \times \cdots \times S_n, \mathcal{S}_1 \otimes \cdots \otimes \mathcal{S}_n)$ and the family is projectively consistent. Then there is a unique probability measure μ on

$$\left(\prod_{n=1}^{\infty} S_n, \bigotimes_{n=1}^{\infty} \mathcal{S}_n\right)$$

with finite-dimensional marginals μ_n.

Proof. Choose Borel isomorphisms from S_n onto Borel subsets $B_n \subseteq [0, 1]$. Transport each finite-dimensional law to $[0, 1]^n$, extending it by zero outside $B_1 \times \cdots \times B_n$. The compact-cube theorem gives a measure on $[0, 1]^{\mathbb{N}}$. Every coordinate belongs to its corresponding B_n with probability one, so the countable product $\prod_n B_n$ has full measure. Restrict to this set and transport the measure back. Uniqueness again follows from the cylinder π-system. $\square$

Theorem 1.34 (Infinite product measures). Let $(\nu_n)_{n \geq 1}$ be Borel probability measures on $\mathbb{R}$. There is a unique probability measure $\nu = \bigotimes_{n=1}^{\infty} \nu_n$ on $(\mathbb{R}^{\mathbb{N}}, \bigotimes_{n \geq 1} \mathcal{B}(\mathbb{R}))$ such that

$$\nu\left(A_1 \times \cdots \times A_n \times \prod_{k>n} \mathbb{R}\right) = \prod_{k=1}^{n} \nu_k(A_k).$$

For every nonnegative measurable, or integrable, function $f : \mathbb{R}^n \to \mathbb{R}$,

$$\int_{\mathbb{R}^{\mathbb{N}}} f(x_1, \ldots, x_n)\, \nu(\mathrm{d}x) = \int_{\mathbb{R}^n} f(x_1, \ldots, x_n) \prod_{k=1}^{n} \nu_k(\mathrm{d}x_k).$$

Proof. Apply the preceding theorem to the consistent finite-dimensional measures $\nu_1 \otimes \cdots \otimes \nu_n$. The integral identity is the finite-dimensional pushforward formula. $\square$

Corollary 1.35 (Construction of an independent sequence). Let (μ_n) be Borel probability measures on $\mathbb{R}$. There is a probability space carrying independent random variables (X_n) such that $\mathcal{L}(X_n) = \mu_n$ for every n. They are identically distributed precisely when all the measures μ_n are equal.

Proof. Take $\Omega = \mathbb{R}^{\mathbb{N}}$ with the product measure $\bigotimes_n \mu_n$, and let $X_n = \pi_n$. The cylinder formula gives both the marginal laws and independence. $\square$

2 Law of Large Numbers

2.1 Convergence in probability

Throughout this section, random variables that agree almost surely are identified.

Definition 2.1 (Convergence in probability). A sequence (X_n) converges to X in probability, written $X_n \xrightarrow{P} X$, if for every $\varepsilon > 0$,

$$P(|X_n - X| > \varepsilon) \longrightarrow 0.$$

Proposition 2.2 (A metric for convergence in probability). On $L^0(\Omega, \mathcal{F}, P)$, define

$$d_0(X, Y) = \mathbb{E}\big[1 \wedge |X - Y|\big].$$

Then d_0 is a metric, and

$$d_0(X_n, X) \to 0 \quad \iff \quad X_n \xrightarrow{P} X.$$

Proof. Only the triangle inequality and the convergence equivalence require comment. Since $1 \wedge (a + b) \leq (1 \wedge a) + (1 \wedge b)$, the triangle inequality follows pointwise. If $d_0(X_n, X) \to 0$, then for $0 < \varepsilon \leq 1$,

$$P(|X_n - X| > \varepsilon) \leq \varepsilon^{-1} d_0(X_n, X).$$

Conversely, convergence in probability implies that $1 \wedge |X_n - X| \to 0$ in probability. Every subsequence has an almost surely convergent further subsequence, and dominated convergence shows that every subsequence of $d_0(X_n, X)$ has a further subsequence tending to zero. Hence the whole numerical sequence tends to zero. □

Theorem 2.3 (Completeness of L^0). The metric space (L^0, d_0) is complete.

Proof. Let (X_n) be Cauchy in probability. Choose a subsequence (X_{n_k}) such that

$$P(|X_{n_{k+1}} - X_{n_k}| > 2^{-k}) \leq 2^{-k}.$$

The first Borel–Cantelli lemma implies that, almost surely, only finitely many events

$$\{|X_{n_{k+1}} - X_{n_k}| > 2^{-k}\}$$

occur. Thus (X_{n_k}) is almost surely Cauchy and converges to a finite random variable X. It follows that $X_{n_k} \to X$ in probability. Since the original sequence is Cauchy in probability, the whole sequence converges to X in probability. □

Proposition 2.4 (Basic properties). Let $X_n \xrightarrow{P} X$ and $Y_n \xrightarrow{P} Y$.

(i) The limit in probability is unique up to almost sure equality.

(ii) $X_n + Y_n \xrightarrow{P} X + Y$ and $X_n Y_n \xrightarrow{P} XY$.

(iii) If $f : \mathbb{R}^d \to \mathbb{R}^m$ is continuous and $Z_n \xrightarrow{P} Z$, then $f(Z_n) \xrightarrow{P} f(Z)$.

(iv) Almost sure convergence implies convergence in probability.

(v) $X_n \xrightarrow{P} X$ if and only if every subsequence has a further subsequence that converges to X almost surely.

Proof. Uniqueness follows from $P(|X - Y| > \varepsilon) \le P(|X - X_n| > \varepsilon/2) + P(|X_n - Y| > \varepsilon/2)$. The sum follows from the same estimate. For products, first note that a sequence converging in probability is bounded in probability, then write $X_n Y_n - XY = X_n(Y_n - Y) + Y(X_n - X)$. The continuous-mapping assertion follows by a subsequence argument, and the last two assertions follow respectively from dominated convergence for indicators and the Borel–Cantelli subsequence construction used above. □

2.2 Independence

Definition 2.5 (Independent families). Families of events $\mathcal{C}_i \subseteq \mathcal{F}$, $i \in I$, are independent if, for every finite set of distinct indices $i_1, \ldots, i_m$ and every $A_j \in \mathcal{C}_{i_j}$,

$$P\left(\bigcap_{j=1}^{m} A_j\right) = \prod_{j=1}^{m} P(A_j).$$

Random variables $(X_i)_{i \in I}$ are independent if the σ-fields $\sigma(X_i)$ are independent.

Definition 2.6 (π-systems and Dynkin systems). A nonempty family $\mathcal{P}$ is a π-system if it is closed under finite intersections. A family $\mathcal{D}$ is a Dynkin system if it contains Ω, is closed under complements, and is closed under countable disjoint unions.

Theorem 2.7 (π–λ theorem). If a Dynkin system $\mathcal{D}$ contains a π-system $\mathcal{P}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{D}$.

Corollary 2.8 (Independence passes to generated σ-fields). If $\mathcal{C}_1, \ldots, \mathcal{C}_m$ are independent π-systems, each containing Ω, then $\sigma(\mathcal{C}_1), \ldots, \sigma(\mathcal{C}_m)$ are independent.

Proof. Fix all but one coordinate and consider the class of sets in the remaining coordinate for which the required product identity holds. It is a Dynkin system containing the corresponding π-system. Apply the π–λ theorem, and repeat one coordinate at a time. □

Lemma 2.9 (Second Borel–Cantelli lemma). If (A_n) is an independent sequence of events and

$\sum_n P(A_n) = \infty$, then

$$P(A_n \text{ i.o.}) = 1.$$

Proof. For $m \le N$, independence and $1 - x \le e^{-x}$ give

$$P\left(\bigcap_{n=m}^{N} A_n^c\right) = \prod_{n=m}^{N} (1 - P(A_n)) \le \exp\left(-\sum_{n=m}^{N} P(A_n)\right).$$

Let $N \to \infty$. The right-hand side tends to zero, so $P(\bigcup_{n \ge m} A_n) = 1$ for every m. Intersect over m. □

Proposition 2.10 (Measurable transformations preserve independence). If $X_1, \ldots, X_m$ are independent and f_i are measurable, then $f_1(X_1), \ldots, f_m(X_m)$ are independent.

Proof. For every Borel set B, the event $\{f_i(X_i) \in B\}$ belongs to $\sigma(X_i)$. □

Theorem 2.11 (Product-law characterization). Random variables $X_i : (\Omega, \mathcal{F}, P) \to (S_i, \mathcal{S}_i)$, $i = 1, \ldots, m$, are independent if and only if

$$\mathcal{L}(X_1, \ldots, X_m) = \mathcal{L}(X_1) \otimes \cdots \otimes \mathcal{L}(X_m).$$

Equivalently, for all bounded measurable f_i,

$$\mathbb{E}\left[\prod_{i=1}^{m} f_i(X_i)\right] = \prod_{i=1}^{m} \mathbb{E}[f_i(X_i)].$$

The same identity holds for nonnegative measurable functions, with values in $[0, \infty]$.

Proof. Independence gives the product identity on measurable rectangles, which form a generating π-system. Uniqueness of product measures gives the law identity. The integral factorization follows first for indicators, then for simple functions, and finally by bounded or monotone convergence. Conversely, apply the functional identity to indicators. □

Proposition 2.12 (Equivalent tests for real-valued independence). For real random variables $X_1, \ldots, X_m$, the following are equivalent:

(i) the variables are independent;

(ii) the bounded-measurable factorization above holds;

(iii) it holds for all bounded continuous f_i;

(iv) it holds for all $f_i \in C_c(\mathbb{R})$ together with constants;

(v) for all $x_1, \ldots, x_m \in \mathbb{R}$,

$$P(X_1 \le x_1, \ldots, X_m \le x_m) = \prod_{i=1}^{m} P(X_i \le x_i).$$

Proof. The forward implications are immediate. The classes of test functions in (iii) and (iv) determine finite Borel measures by the functional monotone-class theorem. The lower orthants in (v) form a π-system that generates $\mathcal{B}(\mathbb{R}^m)$, so equality there determines the joint law. $\square$

Lemma 2.13 (Grouping independent variables). Let $(X_i)_{i\in I}$ be independent, and let $(I_r)_{r\in R}$ be pairwise disjoint subsets of I. Then the σ-fields

$$\mathcal{G}_r = \sigma(X_i : i \in I_r), \qquad r \in R,$$

are independent. Consequently, measurable functions of disjoint groups of independent variables are independent.

Proof. For each r, finite intersections of events from $\{\sigma(X_i) : i \in I_r\}$ form a π-system generating $\mathcal{G}_r$. These π-systems are independent by the definition of independence of the whole family. Apply the preceding π–λ argument. $\square$

Theorem 2.14 (Moment characterization for bounded variables). Let $X_1, \ldots, X_m$ be bounded real random variables. They are independent if and only if, for every $k_1, \ldots, k_m \in \mathbb{N}_0$,

$$\mathbb{E}[X_1^{k_1} \cdots X_m^{k_m}] = \prod_{i=1}^{m} \mathbb{E}[X_i^{k_i}].$$

Proof. Necessity follows from the product-law characterization. Conversely, choose a compact interval containing the ranges of all variables. By linearity, the factorization holds for polynomials in each coordinate. The Weierstrass approximation theorem extends it uniformly to continuous functions on that interval. The bounded-continuous characterization of independence then applies. $\square$

2.3 Weak law of large numbers

Definition 2.15 (Uncorrelated variables). Square-integrable random variables (X_i) are uncorrelated if $\mathrm{Cov}(X_i, X_j) = 0$ whenever $i \neq j$.

Proposition 2.16. If $X_1, \ldots, X_n$ are pairwise uncorrelated, then

$$\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i).$$

Proof. Expand the square of the centered sum. All cross-covariance terms vanish. $\square$

Theorem 2.17 (L^2 weak law). Let (X_n) be pairwise uncorrelated with common mean μ and $\sup_n \mathrm{Var}(X_n) \le C < \infty$. If $S_n = \sum_{k=1}^{n} X_k$, then

$$\frac{S_n}{n} \longrightarrow \mu \quad \text{in } L^2 \text{ and in probability.}$$

Proof. The mean of S_n/n is μ, and

$$\mathbb{E}\left|\frac{S_n}{n} - \mu\right|^2 = \frac{1}{n^2}\sum_{k=1}^{n} \mathrm{Var}(X_k) \le \frac{C}{n} \to 0.$$

□

Proposition 2.18 (Chebyshev form). If S_n is square integrable, $a_n = \mathbb{E}S_n$, and $\mathrm{Var}(S_n)/b_n^2 \to 0$, then

$$\frac{S_n - a_n}{b_n} \xrightarrow{P} 0.$$

Theorem 2.19 (Weak law for triangular arrays). For each n, let $X_{n,1}, \dots, X_{n,n}$ be independent. Let $b_n \to \infty$ and set

$$\overline{X}_{n,k} = X_{n,k}\mathbf{1}_{\{|X_{n,k}| \le b_n\}}, \qquad a_n = \sum_{k=1}^{n} \mathbb{E}\overline{X}_{n,k}.$$

Assume

$$\sum_{k=1}^{n} P(|X_{n,k}| > b_n) \to 0$$

and

$$\frac{1}{b_n^2}\sum_{k=1}^{n} \mathbb{E}[\overline{X}_{n,k}^2] \to 0.$$

Then, with $S_n = \sum_{k=1}^{n} X_{n,k}$,

$$\frac{S_n - a_n}{b_n} \xrightarrow{P} 0.$$

Proof. Let $\overline{S}_n = \sum_k \overline{X}_{n,k}$. The union bound gives

$$P(S_n \ne \overline{S}_n) \le \sum_{k=1}^{n} P(|X_{n,k}| > b_n) \to 0.$$

Independence and Chebyshev's inequality give

$$P(|\overline{S}_n - a_n| > \varepsilon b_n) \le \frac{1}{\varepsilon^2 b_n^2}\sum_{k=1}^{n} \mathrm{Var}(\overline{X}_{n,k}) \le \frac{1}{\varepsilon^2 b_n^2}\sum_{k=1}^{n} \mathbb{E}[\overline{X}_{n,k}^2] \to 0.$$

Combine the two estimates. □

Theorem 2.20 (Feller-type weak law). Let (X_n) be i.i.d. and suppose

$$xP(|X_1| > x) \longrightarrow 0.$$

Put $S_n = \sum_{k=1}^{n} X_k$ and

$$\mu_n = \mathbb{E}[X_1\mathbf{1}_{\{|X_1| \le n\}}].$$

Then

$$\frac{S_n}{n} - \mu_n \xrightarrow{P} 0.$$

Proof. Apply the triangular-array theorem with $X_{n,k} = X_k$ and $b_n = n$. The first condition is $nP(|X_1| > n) \to 0$. For the second, use the layer-cake formula:

$$\frac{1}{n}\mathbb{E}[X_1^2 \mathbf{1}_{\{|X_1| \le n\}}] \le \frac{2}{n}\int_0^n yP(|X_1| > y)\,\mathrm{d}y.$$

The integrand $g(y) = 2yP(|X_1| > y)$ tends to zero and is bounded. Its Cesàro average $n^{-1}\int_0^n g(y)\,\mathrm{d}y$ therefore tends to zero. □

Corollary 2.21 (Khinchin's weak law). If (X_n) is i.i.d. and $\mathbb{E}|X_1| < \infty$, then

$$\frac{S_n}{n} \xrightarrow{P} \mathbb{E}X_1.$$

Proof. We have $xP(|X_1| > x) \le \mathbb{E}[|X_1|\mathbf{1}_{\{|X_1|>x\}}] \to 0$, and dominated convergence gives $\mu_n \to \mathbb{E}X_1$. Apply the preceding theorem. □

2.4 Independent random variables and tail events

Definition 2.22 (Tail σ-field). For a sequence (X_n), the tail σ-field is

$$\mathcal{T} = \bigcap_{m=1}^{\infty} \sigma(X_m, X_{m+1}, \ldots).$$

An event in $\mathcal{T}$ is unaffected by changing finitely many coordinates.

Example. If $b_n \to \infty$, then each event

$$\left\{\lim_{n\to\infty} \frac{S_n}{b_n} = c\right\}$$

is a tail event: changing finitely many summands changes S_n/b_n by a quantity tending to zero. By contrast, an event such as $\{\limsup_n S_n > 0\}$ need not be a tail event, because a finite change can shift every later partial sum by a nonvanishing constant.

Theorem 2.23 (Kolmogorov's zero–one law). If (X_n) is independent, then every $A \in \mathcal{T}$ satisfies $P(A) \in \{0, 1\}$.

Proof. For each m, A is measurable with respect to $\sigma(X_m, X_{m+1}, \ldots)$ and is therefore independent of $\sigma(X_1, \ldots, X_{m-1})$. Hence A is independent of the algebra $\bigcup_m \sigma(X_1, \ldots, X_m)$, and a π–λ argument shows that A is independent of $\sigma(X_1, X_2, \ldots)$. In particular, A is independent of itself, so $P(A) = P(A)^2$. □

Example (Borel–Cantelli zero–one law). For independent events (A_n),

$$P(A_n \text{ i.o.}) = \begin{cases} 0, & \sum_n P(A_n) < \infty, \\ 1, & \sum_n P(A_n) = \infty. \end{cases}$$

The event $\{A_n \text{ i.o.}\}$ is a tail event, and the two cases are the first and second Borel–Cantelli lemmas.

Theorem 2.24 (Approximation by finitely many coordinates). Let Y be bounded and $\sigma(X_1, X_2, \ldots)$-measurable. For every $\varepsilon > 0$, there are $N \in \mathbb{N}$ and a bounded Borel function $F : \mathbb{R}^N \to \mathbb{R}$ such that

$$\mathbb{E}|Y - F(X_1, \ldots, X_N)| < \varepsilon.$$

Proof. The union $\bigcup_N \sigma(X_1, \ldots, X_N)$ is an algebra generating $\sigma(X_1, X_2, \ldots)$. Indicators of sets from this algebra are dense in L^1 among indicators of the generated σ-field, and hence their simple-function span is dense in L^1. Choose a simple approximation measurable with respect to one finite-coordinate σ-field, taking N to be the largest coordinate used. The Doob–Dynkin factorization gives the Borel function F. □

Corollary 2.25 (Tail-measurable variables are constant). If (X_n) is independent and an extended-real random variable Y is $\mathcal{T}$-measurable, then $Y = c$ almost surely for some $c \in [-\infty, \infty]$.

Proof. For every $t \in \mathbb{R}$, the event $\{Y \leq t\}$ is a tail event, so its probability is zero or one. Let $c = \inf\{t \in \mathbb{R} : P(Y \leq t) = 1\}$, allowing infinite values. Monotonicity and right-continuity of the distribution function show that $P(Y = c) = 1$ when c is finite; the cases $c = \pm\infty$ follow in the same way. □

Definition 2.26 (Convolution). For Borel probability measures μ, ν on $\mathbb{R}^d$, their convolution is

$$(\mu * \nu)(B) = \int_{\mathbb{R}^d} \mu(B - y)\, \nu(\mathrm{d}y).$$

If $X \sim \mu$ and $Y \sim \nu$ are independent, then $X + Y \sim \mu * \nu$.

Proof. For Borel B, Tonelli's theorem and the product law give

$$P(X + Y \in B) = \iint \mathbf{1}_B(x + y)\, \mu(\mathrm{d}x)\nu(\mathrm{d}y) = \int \mu(B - y)\, \nu(\mathrm{d}y).$$

□

Proposition 2.27 (Convolution densities). If μ and ν have Lebesgue densities f and g, then $\mu * \nu$ has density

$$(f * g)(u) = \int_{\mathbb{R}^d} f(x)g(u - x)\, \mathrm{d}x.$$

For point masses, the corresponding formula is

$$(\mu * \nu)(\{u\}) = \sum_x \mu(\{x\})\nu(\{u - x\}),$$

where only atoms contribute.

Proof. Use Tonelli's theorem and the translation invariance of Lebesgue measure:

$$(\mu * \nu)(B) = \iint \mathbf{1}_B(x+y)f(x)g(y)\,\mathrm{d}x\mathrm{d}y = \int_B \left(\int f(x)g(u-x)\,\mathrm{d}x \right) \mathrm{d}u.$$

The atomic identity follows directly from the definition. □

2.5 Strong law of large numbers

Definition 2.28 (Tail equivalence). Two sequences (X_n) and (X'_n) on the same probability space are *tail equivalent* if

$$\sum_{n=1}^{\infty} P(X_n \neq X'_n) < \infty.$$

Then $X_n = X'_n$ for all sufficiently large n, almost surely.

Corollary 2.29. If (X_n) and (X'_n) are tail equivalent and $b_n \to \infty$, then

$$\frac{1}{b_n}\sum_{k=1}^{n} X_k \quad \text{and} \quad \frac{1}{b_n}\sum_{k=1}^{n} X'_k$$

have the same almost sure convergence behavior and, when they converge, the same limit.

Proof. Outside a null set, the two sequences differ only in finitely many terms. Their partial sums therefore differ by an eventually constant finite random quantity, which divided by b_n tends to zero. □

Lemma 2.30 (Integrable tails). If $X \in L^1$ and $\varepsilon > 0$, then

$$\sum_{n=1}^{\infty} P(|X| \geq n\varepsilon) \leq \frac{1}{\varepsilon}\mathbb{E}|X|.$$

Consequently, if (X_n) is i.i.d. and integrable, then it is tail equivalent to $X'_n = X_n \mathbf{1}_{\{|X_n| \leq n\}}$.

Proof. For $x \geq 0$, $\sum_{n\geq 1} \mathbf{1}_{[n,\infty)}(x) \leq x$. Apply this to $|X|/\varepsilon$ and take expectations. The consequence follows from the first Borel–Cantelli lemma. □

Theorem 2.31 (Kolmogorov maximal inequality). Let $Y_1, \ldots, Y_n$ be independent, mean-zero, square-integrable random variables, and put $S_k = \sum_{j=1}^{k} Y_j$ and $S_n^* = \max_{k\leq n} |S_k|$. Then, for every $\varepsilon > 0$,

$$\varepsilon^2 P(S_n^* \geq \varepsilon) \leq \mathbb{E}[S_n^2 \mathbf{1}_{\{S_n^* \geq \varepsilon\}}] \leq \mathbb{E}S_n^2.$$

Proof. Let $\tau = \inf\{k : |S_k| \geq \varepsilon\}$ and decompose the event $\{S_n^* \geq \varepsilon\}$ into the disjoint events $\{\tau = j\}$, $j \leq n$. Since $\{\tau = j\} \in \sigma(Y_1, \ldots, Y_j)$ and $S_n - S_j$ is independent of this σ-field with mean zero,

$$\mathbb{E}[S_n^2 \mathbf{1}_{\{\tau=j\}}] = \mathbb{E}[S_j^2 \mathbf{1}_{\{\tau=j\}}] + \mathbb{E}[(S_n - S_j)^2 \mathbf{1}_{\{\tau=j\}}] \geq \varepsilon^2 P(\tau = j).$$

Sum over j. □

Theorem 2.32 (Kolmogorov convergence criterion). Let (Y_n) be independent square-integrable random variables. If

$$\sum_{n=1}^{\infty} \mathrm{Var}(Y_n) < \infty,$$

then

$$\sum_{n=1}^{\infty} (Y_n - \mathbb{E}Y_n)$$

converges almost surely and in L^2. If in addition the numerical series $\sum_n \mathbb{E}Y_n$ converges, then $\sum_n Y_n$ converges almost surely and in L^2.

Proof. The centered partial sums are Cauchy in L^2 by orthogonality. For almost sure convergence, let $S_n = \sum_{k=1}^{n}(Y_k - \mathbb{E}Y_k)$. Kolmogorov's inequality, first on a finite interval and then with the endpoint tending to infinity, gives

$$P\left(\sup_{j\geq m} |S_j - S_m| > \varepsilon\right) \leq \frac{1}{\varepsilon^2} \sum_{j=m+1}^{\infty} \mathrm{Var}(Y_j) \longrightarrow 0.$$

Since

$$\sup_{j,k\geq m} |S_j - S_k| \leq 2 \sup_{j\geq m} |S_j - S_m|,$$

the decreasing random variables on the left converge in probability to zero. Their pointwise limit must therefore vanish almost surely. Thus (S_n) is almost surely Cauchy. Adding a convergent deterministic series proves the final assertion. □

Lemma 2.33 (Kronecker's lemma). Let $b_n > 0$ be nondecreasing with $b_n \to \infty$. If the series $\sum_n x_n / b_n$ converges, then

$$\frac{1}{b_n} \sum_{k=1}^{n} x_k \longrightarrow 0.$$

Proof. Let $s_n = \sum_{k=1}^{n} x_k / b_k$ and $s_n \to s$. Summation by parts gives

$$\frac{1}{b_n} \sum_{k=1}^{n} x_k = s_n - \frac{1}{b_n} \sum_{k=1}^{n-1} (b_{k+1} - b_k) s_k.$$

Subtract and add s. The constant part tends to zero because the weights sum to $(b_n - b_1)/b_n$, while the remainder tends to zero by splitting the weighted sum at a fixed large index and using $s_k \to s$. □

Theorem 2.34 (Kolmogorov strong law). Let (X_n) be i.i.d. with $\mathbb{E}|X_1| < \infty$, and put $S_n = \sum_{k=1}^{n} X_k$. Then

$$\frac{S_n}{n} \longrightarrow \mathbb{E}X_1 \qquad \text{almost surely.}$$

Proof. Let $X_n' = X_n \mathbf{1}_{\{|X_n| \le n\}}$. The integrable-tail lemma shows that (X_n) and (X_n') are tail equivalent. Moreover,

$$\sum_{n=1}^{\infty} \frac{\operatorname{Var}(X_n')}{n^2} \le \mathbb{E}\left[X_1^2 \sum_{n \ge |X_1|} \frac{1}{n^2} \right] \le C\mathbb{E}|X_1| < \infty,$$

where the summand at $X_1 = 0$ is interpreted as zero and the elementary bound $\sum_{n \ge x} n^{-2} \le C/(1+x)$ is used. The convergence criterion and Kronecker's lemma therefore give

$$\frac{1}{n} \sum_{k=1}^{n} (X_k' - \mathbb{E}X_k') \to 0 \quad \text{a.s.}$$

Dominated convergence yields $\mathbb{E}X_k' \to \mathbb{E}X_1$, so Cesàro's lemma gives $n^{-1} \sum_{k \le n} \mathbb{E}X_k' \to \mathbb{E}X_1$. Tail equivalence completes the proof. □

Theorem 2.35 (Marcinkiewicz–Zygmund strong law). Let $1 < p < 2$, let (X_n) be i.i.d., and assume $\mathbb{E}|X_1|^p < \infty$. Then

$$\frac{S_n - n\mathbb{E}X_1}{n^{1/p}} \longrightarrow 0 \qquad \text{almost surely.}$$

Proof. Set $Y_n = X_n \mathbf{1}_{\{|X_n| \le n^{1/p}\}}$. Since

$$\sum_n P(X_n \ne Y_n) = \sum_n P(|X_1|^p > n) \le \mathbb{E}|X_1|^p,$$

the two sequences are tail equivalent. Also,

$$\sum_{n=1}^{\infty} \frac{\operatorname{Var}(Y_n)}{n^{2/p}} \le \mathbb{E}\left[X_1^2 \sum_{n \ge |X_1|^p} n^{-2/p} \right] \le C_p \mathbb{E}|X_1|^p < \infty.$$

The convergence criterion and Kronecker's lemma, with $b_n = n^{1/p}$, imply

$$\frac{1}{n^{1/p}} \sum_{k=1}^{n} (Y_k - \mathbb{E}Y_k) \to 0 \quad \text{a.s.}$$

It remains to control the means. By Tonelli,

$$\frac{1}{n^{1/p}} \sum_{k=1}^{n} \mathbb{E}\left[|X_1| \mathbf{1}_{\{|X_1| > k^{1/p}\}} \right] \le \mathbb{E}\left[\frac{|X_1| \min(n, |X_1|^p)}{n^{1/p}} \right].$$

Split according to $|X_1| \le n^{1/p}$ and its complement. On the first set, the integrand is at most $|X_1|^p$ and tends pointwise to zero; on the second, it is also at most $|X_1|^p$, and its expectation tends to zero. Dominated convergence finishes the mean correction, and tail equivalence finishes the proof. □

Theorem 2.36 (A weighted strong law). Let (X_n) be independent with common mean α and

$\sup_n \operatorname{Var}(X_n) \le s^2$. Let (b_n) be nondecreasing, $b_n \to \infty$, and assume

$$\sum_{n=1}^{\infty} \frac{1}{b_n^2} < \infty.$$

Then

$$\frac{\sum_{k=1}^{n}(X_k - \alpha)}{b_n} \longrightarrow 0 \quad \text{almost surely and in } L^2.$$

Proof. The series $\sum_n (X_n - \alpha)/b_n$ converges almost surely by Kolmogorov's criterion, so Kronecker's lemma gives the almost sure conclusion. For L^2 convergence,

$$\mathbb{E}\left|\frac{\sum_{k=1}^{n}(X_k - \alpha)}{b_n}\right|^2 \le s^2 \frac{n}{b_n^2}.$$

Because b_n is nondecreasing, $a_n = b_n^{-2}$ is nonincreasing; convergence of $\sum_n a_n$ implies $na_n \to 0$. Thus the displayed bound tends to zero. □

2.6 Applications: renewal theory

Theorem 2.37 (Elementary renewal law). Let (X_n) be i.i.d., nonnegative, integrable, and suppose $P(X_1 > 0) > 0$. Put $S_0 = 0$, $S_n = \sum_{k=1}^{n} X_k$, and

$$N_t = \max\{n \ge 0 : S_n \le t\}.$$

Then, with $\mu = \mathbb{E}X_1 > 0$,

$$\frac{N_t}{t} \longrightarrow \frac{1}{\mu} \qquad \text{almost surely as } t \to \infty.$$

Proof. The strong law gives $S_n/n \to \mu > 0$, so $S_n \to \infty$ and $N_t \to \infty$ almost surely. By definition,

$$S_{N_t} \le t < S_{N_t+1}.$$

Divide by N_t and use the strong law along the random subsequences N_t and $N_t + 1$ to obtain $t/N_t \to \mu$. □

Lemma 2.38. If (X_n) is i.i.d. and $X_1 \in L^1$, then

$$\frac{X_n}{n} \longrightarrow 0 \qquad \text{almost surely.}$$

Proof. For every $\varepsilon > 0$, $\sum_n P(|X_n| \ge n\varepsilon) < \infty$ by the integrable-tail lemma. The first Borel–Cantelli lemma gives the result first for rational $\varepsilon > 0$, and then for all ε. □

Theorem 2.39 (Alternating renewal reward). Let $(X_n, Y_n)_{n\ge1}$ be i.i.d. pairs of nonnegative integrable random variables with $P(X_1 + Y_1 > 0) > 0$. Interpret X_n as an "on" time and Y_n as the following "off" time. Put

$$S_n = \sum_{k=1}^{n} (X_k + Y_k), \qquad N_t = \max\{n : S_n \le t\},$$

and

$$T_t = \sum_{k=1}^{N_t} X_k + \min\{X_{N_t+1}, t - S_{N_t}\}.$$

Then

$$\frac{T_t}{t} \longrightarrow \frac{\mathbb{E}X_1}{\mathbb{E}X_1 + \mathbb{E}Y_1} \qquad \text{almost surely.}$$

Proof. Let $\widetilde{T}_t = \sum_{k=1}^{N_t} X_k$. Then $0 \le T_t - \widetilde{T}_t \le X_{N_t+1}$. The preceding lemma and the renewal law imply

$$\frac{X_{N_t+1}}{t} = \frac{X_{N_t+1}}{N_t+1} \frac{N_t+1}{t} \to 0.$$

Moreover, the strong law and renewal law give

$$\frac{\widetilde{T}_t}{t} = \frac{1}{N_t} \sum_{k=1}^{N_t} X_k \cdot \frac{N_t}{t} \longrightarrow \mathbb{E}X_1 \cdot \frac{1}{\mathbb{E}(X_1 + Y_1)}.$$

The squeeze theorem proves the claim. □

2.7 Large deviations

Definition 2.40 (Exponential integrability and cumulants). A random variable X is *exponentially integrable* if $\mathbb{E}e^{t|X|} < \infty$ for every $t > 0$. Its moment-generating function and cumulant-generating function are

$$M_X(t) = \mathbb{E}e^{tX}, \qquad \psi_X(t) = \log M_X(t).$$

They are finite and real analytic on $\mathbb{R}$, and

$$M_X(t) = \sum_{k=0}^{\infty} \mathbb{E}[X^k] \frac{t^k}{k!}.$$

The cumulants are $\kappa_k(X) = \psi_X^{(k)}(0)$; in particular, $\kappa_1(X) = \mathbb{E}X$ and $\kappa_2(X) = \mathrm{Var}(X)$.

Definition 2.41 (Legendre–Fenchel transform). For a finite convex function $\psi : \mathbb{R} \to \mathbb{R}$, define

$$\psi^*(x) = \sup_{t \in \mathbb{R}} \{tx - \psi(t)\} \in (-\infty, \infty].$$

Proposition 2.42. The transform ψ^* is convex and lower semicontinuous, and a finite convex function on $\mathbb{R}$ satisfies $\psi^{**} = \psi$. If $Y = X - \mathbb{E}X$, then $\psi_Y(0) = \psi_Y'(0) = 0$ and $\psi_Y \ge 0$. Hence, for $x \ge 0$,

$$\psi_Y^*(x) = \sup_{t \ge 0} \{tx - \psi_Y(t)\} \ge 0.$$

Proof. A supremum of affine functions is convex and lower semicontinuous. The biconjugacy statement is the supporting-line theorem for finite convex functions. Finally,

$$\psi_Y''(t) = \mathrm{Var}_{Q_t}(Y) \geq 0, \qquad \frac{\mathrm{d}Q_t}{\mathrm{d}P} = e^{tY-\psi_Y(t)},$$

so 0 is a global minimum of ψ_Y. For $t \leq 0$ and $x \geq 0$, $tx - \psi_Y(t) \leq 0$, while $t = 0$ gives zero. □

Theorem 2.43 (One-sided Cramér theorem). Let (X_n) be i.i.d. exponentially integrable random variables. Put $Y_n = X_n - \mathbb{E}X_1$, $S_n = \sum_{k=1}^n Y_k$, and $I = \psi_{Y_1}^*$. Assume Y_1 is not almost surely constant. For every

$$0 < \varepsilon < \operatorname{ess\,sup} Y_1,$$

we have

$$\lim_{n\to\infty} \frac{1}{n} \log P\left(\frac{S_n}{n} \geq \varepsilon\right) = -I(\varepsilon).$$

If $Y_1 = 0$ almost surely, the same interpretation holds with $I(\varepsilon) = \infty$ for $\varepsilon > 0$.

Proof. For every $t \geq 0$, Markov's inequality gives

$$P(S_n \geq n\varepsilon) \leq e^{-tn\varepsilon}\mathbb{E}e^{tS_n} = \exp\{-n(t\varepsilon - \psi_{Y_1}(t))\}.$$

Taking the infimum over t yields the upper bound $\limsup n^{-1} \log P(S_n \geq n\varepsilon) \leq -I(\varepsilon)$.
For the lower bound, convexity and exponential integrability imply that there is $t > 0$ with $\psi_{Y_1}'(t) = \varepsilon$. Tilt the common law of Y_1 by

$$\frac{\mathrm{d}\nu_t}{\mathrm{d}\mathcal{L}(Y_1)}(y) = e^{ty-\psi_{Y_1}(t)},$$

and denote probability and expectation under its n-fold product by $Q_t^{(n)}$ and $\mathbb{E}_t^{(n)}$. Under this product law, the coordinates remain i.i.d., have mean ε, and have finite positive variance. For $\delta > 0$,

$$\begin{aligned} P(n\varepsilon \leq S_n \leq n(\varepsilon+\delta)) &= \mathbb{E}_t^{(n)}\left[e^{-tS_n+n\psi_{Y_1}(t)}\mathbf{1}_{\{n\varepsilon\leq S_n\leq n(\varepsilon+\delta)\}}\right] \\ &\geq e^{-n(t\varepsilon-\psi_{Y_1}(t))-nt\delta}Q_t^{(n)}(n\varepsilon \leq S_n \leq n(\varepsilon+\delta)). \end{aligned}$$

The central limit theorem under the tilted product laws shows that the final probability tends to $1/2$. Since $t\varepsilon - \psi_{Y_1}(t) = I(\varepsilon)$, taking logarithms gives

$$\liminf_{n\to\infty} \frac{1}{n} \log P(S_n \geq n\varepsilon) \geq -I(\varepsilon) - t\delta.$$

Let $\delta \downarrow 0$. □

3 Martingales

3.1 Conditional expectation

Definition 3.1 (Conditioning on an event). Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $B \in \mathcal{F}$ with $P(B) > 0$. The conditional probability measure is

$$P_B(A) = P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad A \in \mathcal{F}.$$

A measurable random variable X belongs to $L^1(P_B)$ precisely when

$$\mathbb{E}[|X|\mathbf{1}_B] < \infty,$$

and then

$$\mathbb{E}_B[X] = \frac{\mathbb{E}[X\mathbf{1}_B]}{P(B)}.$$

In particular, $L^1(P) \subseteq L^1(P_B)$, but equality need not hold.

Definition 3.2 (Conditioning on a countable partition). Let $(A_n)_{n\geq 1}$ be a measurable partition of Ω with $P(A_n) > 0$, and put $\mathcal{G} = \sigma(A_n : n \geq 1)$. For $X \in L^1$,

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_{n=1}^{\infty} \frac{\mathbb{E}[X\mathbf{1}_{A_n}]}{P(A_n)} \mathbf{1}_{A_n}.$$

Theorem 3.3 (Existence and uniqueness of conditional expectation). Let $X \in L^1(\Omega, \mathcal{F}, P)$ and let $\mathcal{G} \subseteq \mathcal{F}$ be a sub-σ-field. There is a $\mathcal{G}$-measurable random variable, unique almost surely, denoted $\mathbb{E}[X \mid \mathcal{G}]$, such that

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A] \qquad (A \in \mathcal{G}).$$

Proof. The signed measure $\nu(A) = \mathbb{E}[X\mathbf{1}_A]$ on $(\Omega, \mathcal{G})$ is absolutely continuous with respect to $P|_{\mathcal{G}}$. Its Radon–Nikodym derivative is a conditional expectation. If Y and Z both satisfy the defining identity, apply it to $\{Y > Z\}$ and $\{Z > Y\}$ to obtain $Y = Z$ almost surely. □

Proposition 3.4 (L^2 projection interpretation). The subspace $L^2(\Omega, \mathcal{G}, P)$ is closed in $L^2(\Omega, \mathcal{F}, P)$. For $X \in L^2$, $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto this subspace. Equivalently, it is the unique $Z \in L^2(\mathcal{G})$ satisfying

$$\mathbb{E}[(X - Z)Y] = 0 \qquad (Y \in L^2(\mathcal{G})).$$

Proof. If $Y_n \in L^2(\mathcal{G})$ and $Y_n \to Y$ in L^2, a subsequence converges almost surely. Its pointwise limit is $\mathcal{G}$-measurable and is a version of Y, so the subspace is closed. The characterization of orthogonal projection gives the second assertion, and taking bounded $\mathcal{G}$-measurable test functions identifies the projection with the conditional expectation defined above. □

Theorem 3.5 (Basic properties). Let $X, Y \in L^1$ and let $\mathcal{G} \subseteq \mathcal{F}$.

(i) Conditional expectation is linear.

(ii) If $X \le Y$ almost surely, then $\mathbb{E}[X \mid \mathcal{G}] \le \mathbb{E}[Y \mid \mathcal{G}]$ almost surely.

(iii) $|\mathbb{E}[X \mid \mathcal{G}]| \le \mathbb{E}[|X| \mid \mathcal{G}]$ almost surely.

(iv) If Z is bounded and $\mathcal{G}$-measurable, then

$$\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}].$$

The same holds whenever both sides are integrable.

(v) If $\mathcal{G}_1 \subseteq \mathcal{G}_2 \subseteq \mathcal{F}$, then

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_2] \mid \mathcal{G}_1] = \mathbb{E}[X \mid \mathcal{G}_1]$$

and

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}_1] \mid \mathcal{G}_2] = \mathbb{E}[X \mid \mathcal{G}_1].$$

(vi) If X is $\mathcal{G}$-measurable, then $\mathbb{E}[X \mid \mathcal{G}] = X$ almost surely.

(vii) $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}X$.

Proof. Linearity and the tower identities follow from the defining integral identity and uniqueness. For monotonicity, let $A = \{\mathbb{E}[X \mid \mathcal{G}] > \mathbb{E}[Y \mid \mathcal{G}]\} \in \mathcal{G}$; integration over A forces $P(A) = 0$. Applying monotonicity to $-|X| \le X \le |X|$ gives (iii). For (iv), test both sides against indicators of sets in $\mathcal{G}$. The remaining assertions are immediate consequences. □

Definition 3.6 (Nonnegative conditional expectation). If $X \ge 0$ is measurable, define

$$\mathbb{E}[X \mid \mathcal{G}] = \lim_{n\to\infty} \mathbb{E}[X \wedge n \mid \mathcal{G}].$$

The limit is $\mathcal{G}$-measurable, may equal $+\infty$, and satisfies the defining integral identity for every $A \in \mathcal{G}$.

Theorem 3.7 (Conditional convergence theorems). Let $\mathcal{G} \subseteq \mathcal{F}$.

(i) If $0 \le X_n \uparrow X$, then

$$\mathbb{E}[X_n \mid \mathcal{G}] \uparrow \mathbb{E}[X \mid \mathcal{G}] \quad \text{almost surely.}$$

(ii) If $X_n \ge 0$, then

$$\mathbb{E}[\liminf_n X_n \mid \mathcal{G}] \le \liminf_n \mathbb{E}[X_n \mid \mathcal{G}] \quad \text{almost surely.}$$

(iii) If $X_n \to X$ almost surely and $|X_n| \le Y \in L^1$, then

$$\mathbb{E}[X_n \mid \mathcal{G}] \to \mathbb{E}[X \mid \mathcal{G}] \quad \text{almost surely and in } L^1.$$

Proof. For (i), the increasing limit has the correct integral over every $A \in \mathcal{G}$ by the ordinary monotone convergence theorem. Part (ii) follows by applying (i) to $Y_k = \inf_{n \geq k} X_n$. For (iii), apply conditional Fatou to $Y \pm X_n$ to identify the almost sure limit. Moreover,

$$\|\mathbb{E}[X_n - X \mid \mathcal{G}]\|_1 \leq \mathbb{E}|X_n - X| \to 0$$

by ordinary dominated convergence. □

Lemma 3.8 (Countable supporting-line representation). If $\varphi : \mathbb{R} \to \mathbb{R}$ is finite and convex, there is a countable family of affine functions (ℓ_j) such that $\ell_j \leq \varphi$ on $\mathbb{R}$ and

$$\varphi(x) = \sup_{j \geq 1} \ell_j(x).$$

Proof. If φ is affine, one line suffices. Otherwise, for each rational q choose a supporting line at q. Convexity makes subgradients locally bounded, so supporting lines based at rationals approaching x take values at x that approach $\varphi(x)$. The countable collection therefore has the stated supremum. □

Theorem 3.9 (Conditional Jensen inequality). Let $X \in L^1$, let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex, and suppose $\varphi(X) \in L^1$. Then

$$\varphi(\mathbb{E}[X \mid \mathcal{G}]) \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}] \quad \text{almost surely.}$$

In particular, for $1 \leq p < \infty$,

$$\|\mathbb{E}[X \mid \mathcal{G}]\|_p \leq \|X\|_p.$$

Proof. For every affine minorant $\ell_j(x) = a_j x + b_j$,

$$\ell_j(\mathbb{E}[X \mid \mathcal{G}]) = \mathbb{E}[\ell_j(X) \mid \mathcal{G}] \leq \mathbb{E}[\varphi(X) \mid \mathcal{G}].$$

Take the countable supremum. An affine lower bound also shows that the negative part of the left-hand side is integrable. The contraction follows from $\varphi(x) = |x|^p$ and integration. □

Proposition 3.10 (Conditioning and independence). Let $X : (\Omega, \mathcal{F}) \to (S, \mathcal{S})$ and let $\mathcal{G} \subseteq \mathcal{F}$. If $\sigma(X)$ and $\mathcal{G}$ are independent, then for every integrable $f(X)$,

$$\mathbb{E}[f(X) \mid \mathcal{G}] = \mathbb{E}f(X) \quad \text{almost surely.}$$

Conversely, if this holds for every bounded measurable f, then $\sigma(X)$ and $\mathcal{G}$ are independent.

Proof. For $A \in \mathcal{G}$, independence gives $\mathbb{E}[f(X)\mathbf{1}_A] = \mathbb{E}f(X)P(A)$, which is the defining identity. Conversely, take $f = \mathbf{1}_B$ and test against $A \in \mathcal{G}$. □

Definition 3.11 (Conditioning on a random variable). For $Y \in L^1$ and a random variable X, write

$$\mathbb{E}[Y \mid X] := \mathbb{E}[Y \mid \sigma(X)].$$

If the state space of X is standard Borel, the Doob–Dynkin lemma gives a measurable function g such that $\mathbb{E}[Y \mid X] = g(X)$ almost surely.

Proposition 3.12 (Independent-parameter substitution). Let X be $\mathcal{G}$-measurable, let Y be independent of $\mathcal{G}$, and let $h : S \times T \to \mathbb{R}$ be measurable with $h(X,Y) \in L^1$. Write $\nu = \mathcal{L}(Y)$ and define

$$g(x) = \begin{cases} \int_T h(x,y)\,\nu(\mathrm{d}y), & \int_T |h(x,y)|\,\nu(\mathrm{d}y) < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

Then g is measurable and

$$\mathbb{E}[h(X,Y) \mid \mathcal{G}] = g(X) \quad \text{almost surely.}$$

Proof. For bounded nonnegative h, the claim is immediate for product indicators $h(x,y) = u(x)v(y)$ and extends by the functional monotone-class theorem. The general nonnegative case follows by monotone convergence. Applying this to $|h|$ shows that the section integral is finite at X almost surely; applying it to h^+ and h^- then proves the integrable case and the measurability of g. □

Example (Conditional density). Suppose (X,Z) has joint density $f_{X,Z}$ and let $f_Z(z) = \int f_{X,Z}(x,z)\,\mathrm{d}x$. Define

$$f_{X|Z}(x \mid z) = \begin{cases} f_{X,Z}(x,z)/f_Z(z), & f_Z(z) > 0, \\ 0, & f_Z(z) = 0. \end{cases}$$

If h is Borel and $\mathbb{E}|h(X)| < \infty$, then

$$\mathbb{E}[h(X) \mid Z] = g(Z), \qquad g(z) = \int h(x) f_{X|Z}(x \mid z)\,\mathrm{d}x.$$

Proof. The function $g(Z)$ is $\sigma(Z)$-measurable. For a Borel set B, Fubini's theorem gives

$$\begin{aligned} \mathbb{E}[g(Z)\mathbf{1}_{\{Z\in B\}}] &= \int_B \int h(x) f_{X,Z}(x,z)\,\mathrm{d}x\mathrm{d}z \\ &= \mathbb{E}[h(X)\mathbf{1}_{\{Z\in B\}}]. \end{aligned}$$

□

3.2 Basics of martingales

Definition 3.13 (Stochastic process and its law). A stochastic process with index set I and state space $(E, \mathcal{E})$ is a family $X = (X_t)_{t \in I}$ of E-valued random variables. Its law is the pushforward of P under the path map

$$\omega \longmapsto (X_t(\omega))_{t \in I}$$

on $(E^I, \mathcal{E}^{\otimes I})$. For Brownian motion with continuous paths, this law is usually viewed on $C([0, \infty))$ and is called Wiener measure.

Definition 3.14 (Common process properties). A real-valued process has independent increments if increments over disjoint time intervals are independent. It is Gaussian if every finite-dimensional marginal is multivariate normal. It is stationary if $(X_{t+s})_t$ and $(X_t)_t$ have the same finite-dimensional distributions for every admissible shift s. It has stationary increments if

$$(X_{t_1+s} - X_s, \dots, X_{t_m+s} - X_s) \overset{d}{=} (X_{t_1} - X_0, \dots, X_{t_m} - X_0)$$

for all admissible $s, t_1, \dots, t_m$.

From this point onward, time is discrete: $n \in \mathbb{N}_0$.

Definition 3.15 (Filtration and adapted process). A filtration is an increasing family $\mathbb{F} = (\mathcal{F}_n)_{n \geq 0}$ of sub-σ-fields of $\mathcal{F}$. A process $X = (X_n)$ is adapted if X_n is $\mathcal{F}_n$-measurable. Its natural filtration is $\mathcal{F}_n^X = \sigma(X_0, \dots, X_n)$.

Definition 3.16 (Martingales). An adapted integrable process $X = (X_n)$ is a martingale, submartingale, or supermartingale according as, for $m \leq n$,

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = X_m, \qquad \mathbb{E}[X_n \mid \mathcal{F}_m] \geq X_m, \qquad \text{or} \qquad \mathbb{E}[X_n \mid \mathcal{F}_m] \leq X_m.$$

Definition 3.17 (Stopping time). A map $\tau : \Omega \to \mathbb{N}_0 \cup \{\infty\}$ is a stopping time if

$$\{\tau \leq n\} \in \mathcal{F}_n \qquad (n \geq 0).$$

Equivalently, $\{\tau = n\} \in \mathcal{F}_n$ for every n.

Proposition 3.18 (Operations on stopping times). If σ and τ are stopping times, then $\sigma \wedge \tau$, $\sigma \vee \tau$, and $\sigma + \tau$ are stopping times. For every deterministic $r \in \mathbb{N}_0$, $\tau + r$ is a stopping time. In general, $\tau - r$ need not be a stopping time.

Proof. Use

$$\{\sigma \wedge \tau \leq n\} = \{\sigma \leq n\} \cup \{\tau \leq n\}, \qquad \{\sigma \vee \tau \leq n\} = \{\sigma \leq n\} \cap \{\tau \leq n\},$$

and

$$\{\sigma + \tau \le n\} = \bigcup_{k=0}^{n} \{\sigma = k\} \cap \{\tau \le n - k\}.$$

Every term in the final finite union belongs to $\mathcal{F}_n$. □

Definition 3.19 (Stopping-time σ-field). For a stopping time τ, define

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \le n\} \in \mathcal{F}_n \text{ for every } n\}.$$

Proposition 3.20. If $\sigma \le \tau$, then $\mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$. Moreover,

$$\mathcal{F}_{\sigma \wedge \tau} = \mathcal{F}_\sigma \cap \mathcal{F}_\tau.$$

If X is adapted and $\tau < \infty$ almost surely, then X_τ is $\mathcal{F}_\tau$-measurable.

Proof. For the first inclusion, if $A \in \mathcal{F}_\sigma$, then

$$A \cap \{\tau \le n\} = (A \cap \{\sigma \le n\}) \cap \{\tau \le n\} \in \mathcal{F}_n.$$

The equality for the minimum follows from $\{\sigma \wedge \tau \le n\} = \{\sigma \le n\} \cup \{\tau \le n\}$. Finally, for Borel B,

$$\{X_\tau \in B\} \cap \{\tau \le n\} = \bigcup_{k=0}^{n} \{\tau = k\} \cap \{X_k \in B\} \in \mathcal{F}_n.$$

□

Proposition 3.21 (Elementary closure properties).

(i) X is a supermartingale if and only if $-X$ is a submartingale.

(ii) Linear combinations of martingales are martingales.

(iii) Nonnegative linear combinations of submartingales, respectively supermartingales, stay in the same class.

(iv) The maximum of two submartingales is a submartingale, and the minimum of two supermartingales is a supermartingale.

(v) If X is a supermartingale and $\mathbb{E}X_N \ge \mathbb{E}X_0$ for some N, then $(X_n)_{0 \le n \le N}$ is a martingale.

Proof. Only (iv) needs a calculation. Using $x \vee y = (x + y + |x - y|)/2$ and conditional Jensen for the absolute-value function gives

$$\mathbb{E}[X_n \vee Y_n \mid \mathcal{F}_m] \ge \mathbb{E}[X_n \mid \mathcal{F}_m] \vee \mathbb{E}[Y_n \mid \mathcal{F}_m] \ge X_m \vee Y_m.$$

The supermartingale statement follows by negation. For (v), expectations of a supermartingale are nonincreasing. Equality at the endpoints forces equality at every intermediate step, and the nonpositive conditional drifts must vanish. □

Theorem 3.22 (Convex transforms). Let X be a martingale and let $\varphi : \mathbb{R} \to \mathbb{R}$ be convex. If $\varphi(X_n)$ is integrable for every n, then $(\varphi(X_n))$ is a submartingale. More generally, if X is a submartingale and φ is increasing and convex, then $(\varphi(X_n))$ is a submartingale whenever it is integrable. Consequently,

$$|X_n|^p \quad (p \geq 1), \qquad (X_n - a)^+,$$

are submartingales in the applicable cases, and if X is a supermartingale, then $X_n \wedge a$ is a supermartingale.

Proof. Conditional Jensen gives

$$\mathbb{E}[\varphi(X_n) \mid \mathcal{F}_m] \geq \varphi(\mathbb{E}[X_n \mid \mathcal{F}_m]).$$

For a martingale this equals $\varphi(X_m)$; for a submartingale, monotonicity of φ makes it at least $\varphi(X_m)$. Finally, $X \wedge a = a - (a - X)^+$ and $-X$ is a submartingale when X is a supermartingale. □

3.3 Optional sampling and stochastic integration

Definition 3.23 (Predictable process). A discrete-time process $H = (H_n)_{n \geq 1}$ is predictable if H_n is $\mathcal{F}_{n-1}$-measurable for every n.

Theorem 3.24 (Doob decomposition). Every adapted integrable process $X = (X_n)$ has a unique decomposition

$$X_n = M_n + A_n,$$

where M is a martingale, $A_0 = 0$, and A is predictable. Explicitly,

$$A_n = \sum_{k=1}^{n} \left(\mathbb{E}[X_k \mid \mathcal{F}_{k-1}] - X_{k-1}\right), \qquad M_n = X_n - A_n.$$

The process X is a submartingale if and only if A is increasing, and it is a supermartingale if and only if A is decreasing.

Proof. The increment $M_k - M_{k-1} = X_k - \mathbb{E}[X_k \mid \mathcal{F}_{k-1}]$ has conditional mean zero, so M is a martingale. The assertions about monotonicity follow straight from the increments of A. If $X = M + A = M' + A'$, then $M - M' = A' - A$ is a predictable martingale starting at zero. Its increments are both $\mathcal{F}_{n-1}$-measurable and have conditional mean zero, hence vanish almost surely. This proves uniqueness. □

Definition 3.25 (Predictable quadratic variation). For a square-integrable martingale M, its predictable quadratic variation is

$$\langle M \rangle_n = \sum_{k=1}^{n} \mathbb{E}[(M_k - M_{k-1})^2 \mid \mathcal{F}_{k-1}].$$

It is the unique predictable increasing process starting at zero such that $M_n^2 - \langle M \rangle_n$ is a martingale.

Proposition 3.26. For every square-integrable martingale M,

$$\mathbb{E}\langle M \rangle_n = \mathbb{E}[(M_n - M_0)^2]$$

and

$$\mathbb{E}M_n^2 = \mathbb{E}M_0^2 + \mathbb{E}\langle M \rangle_n.$$

Proof. Martingale differences are pairwise orthogonal in L^2 and are orthogonal to M_0. Summing their second moments gives both identities. □

Theorem 3.27 (Optional sampling for bounded stopping times). Let X be a supermartingale and let $\sigma \leq \tau \leq N$ be stopping times. Then

$$\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] \leq X_\sigma \quad \text{almost surely,}$$

and therefore $\mathbb{E}X_\tau \leq \mathbb{E}X_\sigma$. For a martingale, equality holds. Conversely, an adapted integrable process X is a martingale if and only if

$$\mathbb{E}X_\tau = \mathbb{E}X_0$$

for every bounded stopping time τ.

Proof. For $A \in \mathcal{F}_\sigma$,

$$X_\tau - X_\sigma = \sum_{k=1}^{N} \mathbf{1}_{\{\sigma < k \leq \tau\}}(X_k - X_{k-1}).$$

The factor $\mathbf{1}_A \mathbf{1}_{\{\sigma<k\leq\tau\}}$ is $\mathcal{F}_{k-1}$-measurable. Taking expectations shows $\mathbb{E}[(X_\tau - X_\sigma)\mathbf{1}_A] \leq 0$, which is the conditional inequality. For the converse, fix $m < n$ and $A \in \mathcal{F}_m$. Apply the assumption to the stopping time equal to m on A and n on A^c, and compare it with the deterministic stopping time n. This gives $\mathbb{E}[X_m \mathbf{1}_A] = \mathbb{E}[X_n \mathbf{1}_A]$. □

Corollary 3.28 (Nonnegative supermartingales). Let X be a nonnegative supermartingale and let $\sigma \leq \tau < \infty$ almost surely. Then $X_\sigma, X_\tau \in L^1$ and

$$\mathbb{E}[X_\tau \mid \mathcal{F}_\sigma] \leq X_\sigma.$$

Proof. Optional sampling applied to $\tau \wedge n$ gives $\mathbb{E}X_{\tau\wedge n} \le \mathbb{E}X_0$. Fatou's lemma gives $\mathbb{E}X_\tau \le \mathbb{E}X_0$, and similarly for σ. For $A \in \mathcal{F}_\sigma$, apply bounded optional sampling to $\sigma \wedge n$ and $\tau \wedge n$ and integrate over $A \cap \{\sigma \le n\} \in \mathcal{F}_{\sigma\wedge n}$. This gives

$$\mathbb{E}[X_{\tau\wedge n}\mathbf{1}_{A\cap\{\sigma\le n\}}] \le \mathbb{E}[X_\sigma \mathbf{1}_{A\cap\{\sigma\le n\}}].$$

Fatou's lemma on the left and monotone convergence on the right yield $\mathbb{E}[X_\tau \mathbf{1}_A] \le \mathbb{E}[X_\sigma \mathbf{1}_A]$. □

Definition 3.29 (Stopped process). For a stopping time τ, define

$$X_n^\tau = X_{n\wedge\tau}, \qquad \mathcal{F}_n^\tau = \mathcal{F}_{n\wedge\tau}.$$

Theorem 3.30 (Stopping preserves martingales). If X is a martingale, submartingale, or supermartingale, then X^τ has the same property with respect to both $\mathbb{F}$ and the stopped filtration $\mathbb{F}^\tau$.

Proof. For a submartingale,

$$X_n^\tau - X_{n-1}^\tau = \mathbf{1}_{\{\tau\ge n\}}(X_n - X_{n-1}),$$

and the indicator is $\mathcal{F}_{n-1}$-measurable. Conditional expectation therefore gives a nonnegative drift. Conditioning once more on the smaller σ-field $\mathcal{F}_{n-1}^\tau$ proves the assertion for the stopped filtration. The other cases follow similarly or by negation. □

Definition 3.31 (Discrete stochastic integral). Let X be adapted and let H be predictable. Whenever the summands are integrable, define

$$(H \cdot X)_n = \sum_{k=1}^{n} H_k(X_k - X_{k-1}).$$

Theorem 3.32 (Stability under bounded predictable transforms). Let X be adapted and integrable.

(i) X is a martingale if and only if $H \cdot X$ is a martingale for every bounded predictable process H.

(ii) X is a submartingale, respectively supermartingale, if and only if $H \cdot X$ has the same property for every bounded nonnegative predictable H.

Proof. If X is a martingale, then

$$\mathbb{E}[H_n(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}] = 0.$$

The sign version is identical. Conversely, choose $H_k = \mathbf{1}_{\{k=n\}}$ to isolate each one-step increment. □

Remark. Taking $H_k = \mathbf{1}_{\{\tau\ge k\}}$ gives

$$X_{n\wedge\tau} = X_0 + (H \cdot X)_n,$$

which is another proof that stopping preserves the martingale class.

3.4 Convergence theorems

3.4.1 Almost sure convergence

Definition 3.33 (Upcrossings). For a real sequence $x = (x_n)$ and $a < b$, define successively

$$S_1 = \inf\{n : x_n \le a\}, \qquad T_1 = \inf\{n \ge S_1 : x_n \ge b\},$$

and

$$S_{k+1} = \inf\{n \ge T_k : x_n \le a\}, \qquad T_{k+1} = \inf\{n \ge S_{k+1} : x_n \ge b\},$$

with $\inf \varnothing = \infty$. The number of completed upcrossings by time n is

$$U_n[a,b](x) = \sum_{k \ge 1} \mathbf{1}_{\{T_k \le n\}},$$

and $U_\infty[a,b] = \lim_n U_n[a,b]$.

Lemma 3.34 (Deterministic upcrossing criterion). A real sequence (x_n) converges in $[-\infty, \infty]$ if and only if $U_\infty[a,b](x) < \infty$ for every pair of rational numbers $a < b$.

Proof. A convergent sequence can cross a fixed interval only finitely many times. Conversely, if $\liminf x_n < \limsup x_n$, choose rationals $\liminf x_n < a < b < \limsup x_n$. The sequence then completes infinitely many upcrossings of $[a,b]$. □

Theorem 3.35 (Doob upcrossing inequality). If $X = (X_n)$ is a submartingale, then for every $a < b$ and n,

$$(b-a)\mathbb{E}U_n[a,b](X) \le \mathbb{E}[(X_n - a)^+] - \mathbb{E}[(X_0 - a)^+].$$

Proof. Let $Y_n = (X_n - a)^+$, a submartingale. Define the predictable process

$$H_j = \sum_{k \ge 1} \mathbf{1}_{\{S_k < j \le T_k\}},$$

which takes values in $\{0, 1\}$. The gain from every completed transaction is at least $b - a$, and an unfinished transaction has nonnegative value, so

$$(H \cdot Y)_n \ge (b-a)U_n[a,b].$$

The process $K = 1 - H$ is also nonnegative and predictable, hence $K \cdot Y$ is a submartingale starting at zero. Since

$$(H \cdot Y)_n + (K \cdot Y)_n = Y_n - Y_0,$$

we have $\mathbb{E}(H \cdot Y)_n \le \mathbb{E}(Y_n - Y_0)$, proving the result. □

Theorem 3.36 (Martingale convergence theorem). If a submartingale or supermartingale (X_n) is bounded in L^1, then there is $X_\infty \in L^1$ such that

$$X_n \longrightarrow X_\infty \qquad \text{almost surely.}$$

Proof. It is enough to treat a submartingale. For rational $a < b$, the upcrossing inequality and the L^1 bound give $\sup_n \mathbb{E}U_n[a,b] < \infty$. Monotone convergence implies $\mathbb{E}U_\infty[a,b] < \infty$, so the number of upcrossings is finite almost surely. Intersecting over the countable set of rational pairs and applying the deterministic criterion gives an almost sure limit in the extended reals. Fatou's lemma gives

$$\mathbb{E}|X_\infty| \le \liminf_n \mathbb{E}|X_n| < \infty,$$

so the limit is finite almost surely. □

Corollary 3.37. Every nonnegative supermartingale converges almost surely to some $X_\infty \in L^1$, and

$$X_n \ge \mathbb{E}[X_\infty \mid \mathcal{F}_n] \qquad (n \ge 0).$$

Proof. The family is L^1-bounded because $\mathbb{E}X_n \le \mathbb{E}X_0$. For $m \ge n$, $X_n \ge \mathbb{E}[X_m \mid \mathcal{F}_n]$. Conditional Fatou gives

$$\mathbb{E}[X_\infty \mid \mathcal{F}_n] \le \liminf_{m\to\infty} \mathbb{E}[X_m \mid \mathcal{F}_n] \le X_n.$$

□

3.4.2 Uniform integrability and L^1 convergence

Definition 3.38 (Uniform integrability). A family $\mathcal{X} \subset L^1(P)$ is uniformly integrable if

$$\lim_{K\to\infty} \sup_{X\in\mathcal{X}} \mathbb{E}[|X|\mathbf{1}_{\{|X|>K\}}] = 0.$$

Theorem 3.39 (de la Vallée–Poussin criterion). A family $\mathcal{X} \subset L^1(P)$ is uniformly integrable if and only if there is an increasing convex function $\Phi : [0,\infty) \to [0,\infty)$ such that

$$\frac{\Phi(x)}{x} \longrightarrow \infty \quad \text{and} \quad \sup_{X\in\mathcal{X}} \mathbb{E}\Phi(|X|) < \infty.$$

Proof. If such Φ exists, let $c_K = \inf_{x\ge K} \Phi(x)/x \to \infty$. Then

$$\mathbb{E}[|X|\mathbf{1}_{\{|X|\ge K\}}] \le c_K^{-1}\mathbb{E}\Phi(|X|),$$

uniformly in X. Conversely, choose $a_n \uparrow \infty$ so that

$$\sup_{X\in\mathcal{X}} \mathbb{E}[(|X| - a_n)^+] < 2^{-n},$$

and put $\Phi(x) = \sum_{n\ge1}(x - a_n)^+$. This is increasing and convex, its ratio to x tends to infinity, and monotone convergence gives $\sup_X \mathbb{E}\Phi(|X|) \le 1$. $\square$

Corollary 3.40. Every L^p-bounded family with $p > 1$ is uniformly integrable. In particular, a family with uniformly bounded means in absolute value and uniformly bounded variances is uniformly integrable.

Proof. Use $\Phi(x) = x^p$. In the second assertion, $\mathbb{E}X^2 = (\mathbb{E}X)^2 + \mathrm{Var}(X)$ is uniformly bounded. $\square$

Proposition 3.41 (Elementary stability). Finite subsets of L^1 are uniformly integrable. If $\mathcal{X}$ and $\mathcal{Y}$ are uniformly integrable, then so are $\mathcal{X} + \mathcal{Y}$, $\mathcal{X} - \mathcal{Y}$, and $\{|X| : X \in \mathcal{X}\}$. If each Y in a family $\mathcal{Y}$ satisfies $|Y| \le |X|$ for some $X \in \mathcal{X}$, then uniform integrability of $\mathcal{X}$ implies that of $\mathcal{Y}$.

Theorem 3.42 (Vitali convergence theorem). For random variables (X_n) and X, the following are equivalent:

(i) $X_n \to X$ in L^1;

(ii) (X_n) is Cauchy in L^1 and $X_n \to X$ in probability;

(iii) (X_n) is uniformly integrable and $X_n \to X$ in probability.

Proof. Clearly (i) implies (ii). If (ii) holds, completeness of L^1 gives an L^1 limit Y. Then $X_n \to Y$ in probability, so uniqueness of limits in probability gives $X = Y$ almost surely and proves (i).
If (i) holds, convergence in probability follows. To prove uniform integrability, choose N so that $\|X_n - X\|_1 < \varepsilon$ for $n \ge N$. For the tail of the sequence,

$$\mathbb{E}[|X_n|\mathbf{1}_{\{|X_n|>K\}}] \le \varepsilon + \mathbb{E}[|X|\mathbf{1}_{\{|X_n|>K\}}].$$

The expectations $\mathbb{E}|X_n|$ are uniformly bounded for $n \ge N$, so Markov's inequality makes the probabilities of the displayed events uniformly small as $K \to \infty$. Absolute continuity of the integral of $|X|$ then controls the second term; the finitely many earlier variables cause no problem. Thus (i) implies (iii).
Conversely, assume (iii). Uniform integrability implies $\sup_n \mathbb{E}|X_n| < \infty$. An almost surely convergent subsequence and Fatou's lemma show that $X \in L^1$. For $K > 0$,

$$\begin{aligned}\mathbb{E}|X_n - X| \le{}& \mathbb{E}(|X_n - X| \wedge K) \\ &+ 2\mathbb{E}[|X_n|\mathbf{1}_{\{|X_n|>K/2\}}] + 2\mathbb{E}[|X|\mathbf{1}_{\{|X|>K/2\}}].\end{aligned}$$

For fixed K, the first term tends to zero because the bounded variables $|X_n - X| \wedge K$ converge to zero in probability. The last two terms are uniformly small for large K, proving L^1 convergence. $\square$

Lemma 3.43 (Conditional expectations of one variable are uniformly integrable). For every $Z \in L^1$, the family

$$\{\mathbb{E}[Z \mid \mathcal{G}] : \mathcal{G} \subseteq \mathcal{F} \text{ a sub-}\sigma\text{-field}\}$$

is uniformly integrable.

Proof. Put $Y = \mathbb{E}[Z \mid \mathcal{G}]$ and $A = \{|Y| > K\} \in \mathcal{G}$. Then

$$KP(A) \leq \mathbb{E}[|Y|\mathbf{1}_A] \leq \mathbb{E}[|Z|\mathbf{1}_A].$$

Also $P(A) \leq \mathbb{E}|Z|/K$. Absolute continuity of the integral of $|Z|$ therefore makes the final term uniformly small as $K \to \infty$. □

Theorem 3.44 (L^1 martingale convergence). For a martingale (X_n), the following are equivalent:

(i) $X_n \to X_\infty$ almost surely and in L^1;

(ii) there is $Z \in L^1$ such that $X_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ for every n;

(iii) the family (X_n) is uniformly integrable.

When these conditions hold, one may take $Z = X_\infty$, and

$$X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n].$$

Proof. If (i) holds, then for $m \geq n$, $X_n = \mathbb{E}[X_m \mid \mathcal{F}_n]$. Contractivity of conditional expectation allows $m \to \infty$ in L^1, giving (ii) with $Z = X_\infty$. Condition (ii) implies (iii) by the preceding lemma. If (iii) holds, the family is L^1-bounded, so the martingale convergence theorem gives an almost sure limit. Vitali's theorem upgrades the convergence to L^1. □

Theorem 3.45 (Optional sampling for uniformly integrable martingales). Let (X_n) be a uniformly integrable martingale with L^1 limit X_∞. For a stopping time T, define $X_T = X_\infty$ on $\{T = \infty\}$. Then

$$X_T = \mathbb{E}[X_\infty \mid \mathcal{F}_T].$$

Consequently, $X_T \in L^1$, the family of all stopped values (X_T) is uniformly integrable, and if $S \leq T$, then

$$X_S = \mathbb{E}[X_T \mid \mathcal{F}_S].$$

Proof. We have $X_n = \mathbb{E}[X_\infty \mid \mathcal{F}_n]$. Conditional Jensen gives

$$\mathbb{E}|X_T| \leq \sum_{n\geq 0} \mathbb{E}[|X_\infty|\mathbf{1}_{\{T=n\}}] + \mathbb{E}[|X_\infty|\mathbf{1}_{\{T=\infty\}}] = \mathbb{E}|X_\infty|.$$

For $A \in \mathcal{F}_T$, decompose according to $\{T = n\}$ and $\{T = \infty\}$ to obtain $\mathbb{E}[X_T\mathbf{1}_A] = \mathbb{E}[X_\infty\mathbf{1}_A]$. This proves the conditional-expectation identity. Uniform integrability follows from the preceding lemma, and the final assertion follows from the tower property and $\mathcal{F}_S \subseteq \mathcal{F}_T$. □

Theorem 3.46 (Optional sampling for nonnegative or uniformly integrable supermartingales). Let X be a supermartingale and suppose either

(a) $X_n \geq 0$ for every n, or

(b) (X_n) is uniformly integrable.

If $S \leq T < \infty$ almost surely, then $X_S, X_T \in L^1$ and

$$\mathbb{E}[X_T \mid \mathcal{F}_S] \leq X_S.$$

Proof. Case (a) was proved above. For case (b), write the Doob decomposition as $X = M - A$, where A is predictable, increasing, and $A_0 = 0$. Since

$$\mathbb{E}A_n = \mathbb{E}X_0 - \mathbb{E}X_n$$

is bounded, $A_n \uparrow A_\infty$ in L^1. Thus (A_n) is uniformly integrable, and so is the martingale $M = X + A$. The martingale is therefore closed, and all its stopped values are uniformly integrable; also $0 \leq A_T \leq A_\infty$. Hence $X_{T\wedge n} \to X_T$ and $X_{S\wedge n} \to X_S$ in L^1. For $B \in \mathcal{F}_S$, integrate bounded optional sampling over $B \cap \{S \leq n\} \in \mathcal{F}_{S\wedge n}$ and then let $n \to \infty$. The L^1 convergences and $\mathbf{1}_{B\cap\{S\leq n\}} \to \mathbf{1}_B$ give

$$\mathbb{E}[X_T \mathbf{1}_B] \leq \mathbb{E}[X_S \mathbf{1}_B],$$

which is the asserted conditional inequality. $\square$

Example (Gambler's ruin). Let (S_n) be simple symmetric random walk with $S_0 = 0$, and let $a < 0 < b$ be integers. Define

$$\tau = \inf\{n \geq 0 : S_n \in \{a, b\}\}.$$

Then $\tau < \infty$ almost surely and

$$P(S_\tau = a) = \frac{b}{b-a}, \qquad P(S_\tau = b) = \frac{-a}{b-a}.$$

Indeed, from every interior state there is a probability at least $2^{-(b-a)}$ of hitting a boundary within the next $b - a$ steps. Iterating over blocks shows that $P(\tau > m(b-a)) \to 0$. The stopped martingale is bounded, so optional sampling gives $0 = \mathbb{E}S_\tau$, and the two displayed probabilities follow from their sum being one.

Example (Expected exit time). For the same stopping time, $S_n^2 - n$ is a martingale. Bounded optional sampling gives

$$\mathbb{E}(\tau \wedge n) = \mathbb{E}S^2_{\tau\wedge n} \leq \max\{a^2, b^2\}.$$

Monotone convergence therefore gives $\mathbb{E}\tau < \infty$, while dominated convergence applies to the bounded stopped positions. Hence

$$\mathbb{E}\tau = \mathbb{E}S_\tau^2 = a^2\frac{b}{b-a} + b^2\frac{-a}{b-a} = |a|b.$$

If τ_a is the first hitting time of a fixed level $a < 0$, then the events that a is hit before b increase to $\{\tau_a < \infty\}$ as $b \to \infty$. Their probabilities tend to one by the gambler's-ruin formula, so $\tau_a < \infty$ almost surely. Moreover, the exit times increase to τ_a, and their expectations $|a|b$ tend to infinity. Thus $\mathbb{E}\tau_a = \infty$.

3.4.3 L^p convergence for $p > 1$

Lemma 3.47 (Doob maximal inequality, weak form). If X is a submartingale, set

$$X_n^* = \max_{0\le k\le n} X_k.$$

Then, for every $\lambda > 0$,

$$\lambda P(X_n^* \ge \lambda) \le \mathbb{E}[X_n \mathbf{1}_{\{X_n^* \ge \lambda\}}].$$

Proof. Let $\tau = \inf\{k : X_k \ge \lambda\} \wedge n$. Bounded optional sampling gives $\mathbb{E}X_\tau \le \mathbb{E}X_n$. On the event $\{X_n^* < \lambda\}$, $X_\tau = X_n$, while on its complement $X_\tau \ge \lambda$. Rearranging yields the inequality. □

Theorem 3.48 (Doob's L^p inequalities). Let X be a martingale, or a nonnegative submartingale, and put $|X|_n^* = \max_{k\le n}|X_k|$.

(i) For $p \ge 1$ and $\lambda > 0$,

$$\lambda^p P(|X|_n^* \ge \lambda) \le \mathbb{E}|X_n|^p.$$

(ii) For $p > 1$,

$$\mathbb{E}|X_n|^p \le \mathbb{E}(|X|_n^*)^p \le \left(\frac{p}{p-1}\right)^p \mathbb{E}|X_n|^p.$$

Proof. The process $|X|^p$ is a nonnegative submartingale, so the weak inequality follows from the lemma. For the strong inequality, apply the weak form to $|X|$ and use the layer-cake formula. For $K > 0$,

$$\begin{aligned}
\mathbb{E}[(|X|_n^* \wedge K)^p] &= \int_0^K p\lambda^{p-1} P(|X|_n^* \ge \lambda)\,\mathrm{d}\lambda \\
&\le p\mathbb{E}\left[|X_n| \int_0^{|X|_n^*\wedge K} \lambda^{p-2}\,\mathrm{d}\lambda\right] \\
&= \frac{p}{p-1}\mathbb{E}[|X_n|(|X|_n^* \wedge K)^{p-1}].
\end{aligned}$$

Hölder's inequality gives the stated constant. Let $K \to \infty$. □

Theorem 3.49 (L^p martingale convergence). Let $p > 1$ and let (X_n) be a martingale satisfying

$$\sup_n \mathbb{E}|X_n|^p < \infty.$$

With $\mathcal{F}_\infty = \sigma(\bigcup_n \mathcal{F}_n)$, there is $X_\infty \in L^p(\mathcal{F}_\infty)$ such that

$$X_n \to X_\infty \quad \text{almost surely and in } L^p.$$

Moreover, $(|X_n|^p)$ is uniformly integrable.

Proof. The L^p bound implies uniform integrability, so an almost sure and L^1 limit exists. Doob's inequality and monotone convergence give

$$\mathbb{E} \sup_{n \geq 0} |X_n|^p \leq \left(\frac{p}{p-1}\right)^p \sup_n \mathbb{E}|X_n|^p < \infty.$$

Thus $|X_n - X_\infty|^p$ is dominated by an integrable multiple of $\sup_k |X_k|^p$, and dominated convergence gives L^p convergence. The same dominator proves uniform integrability of $(|X_n|^p)$. □

Corollary 3.50 (Square-integrable criterion). For a square-integrable martingale M, the following are equivalent:

(i) $\sup_n \mathbb{E} M_n^2 < \infty$;

(ii) $\sup_n \mathbb{E}\langle M \rangle_n < \infty$;

(iii) M_n converges in L^2;

(iv) M_n converges almost surely and in L^2.

Proof. The identity $\mathbb{E} M_n^2 = \mathbb{E} M_0^2 + \mathbb{E}\langle M \rangle_n$ gives (i)⇔(ii). The L^2 martingale convergence theorem gives (i)⇒(iv), while (iv)⇒(iii)⇒(i) is immediate. □

Theorem 3.51 (Almost sure convergence from finite quadratic variation). If M is square integrable and

$$\sup_n \langle M \rangle_n < \infty \quad \text{almost surely},$$

then M_n converges almost surely.

Proof. Assume $M_0 = 0$. For $K > 0$, set

$$\tau_K = \inf\{n : \langle M \rangle_{n+1} > K\}.$$

Predictability makes τ_K a stopping time, and $\langle M^{\tau_K} \rangle_n \leq K$. The square-integrable criterion shows that M^{τ_K} converges almost surely. Since $P(\tau_K = \infty) \to 1$ as $K \to \infty$, the original martingale converges on a full-probability union of these events. □

3.4.4 Applications of martingale convergence

Theorem 3.52 (Lévy's upward theorem). Let $X \in L^1$ and let $\mathcal{F}_n \uparrow \mathcal{F}_\infty := \sigma(\bigcup_n \mathcal{F}_n)$. Then

$$\mathbb{E}[X \mid \mathcal{F}_n] \longrightarrow \mathbb{E}[X \mid \mathcal{F}_\infty] \quad \text{almost surely and in } L^1.$$

Proof. The process $Y_n = \mathbb{E}[X \mid \mathcal{F}_n]$ is a closed, hence uniformly integrable, martingale. Let Y_∞ be its L^1 limit. For $A \in \bigcup_n \mathcal{F}_n$, $\mathbb{E}[Y_\infty \mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A]$. A π–λ argument extends the identity to all $A \in \mathcal{F}_\infty$, identifying $Y_\infty = \mathbb{E}[X \mid \mathcal{F}_\infty]$. □

Corollary 3.53 (Lévy's zero–one form). If $A \in \mathcal{F}_\infty$, then

$$\mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_n] \to \mathbf{1}_A \quad \text{almost surely and in } L^1.$$

Remark. For independent (X_n) and $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, a tail event A is independent of each $\mathcal{F}_n$, so $\mathbb{E}[\mathbf{1}_A \mid \mathcal{F}_n] = P(A)$. Lévy's theorem therefore gives $\mathbf{1}_A = P(A)$ almost surely, recovering Kolmogorov's zero–one law.

3.5 Reverse martingales and exchangeability

Definition 3.54 (Exchangeability). Let E be a standard Borel space. A sequence $(X_n)_{n\geq 1}$ of E-valued random variables is exchangeable if its joint law is invariant under every finite permutation of the coordinates.

Let S_n denote the permutations of $\mathbb{N}$ that fix every index larger than n. On the canonical space $E^{\mathbb{N}}$, let $\mathcal{E}'_n$ be the σ-field of sets invariant under every permutation in S_n, and set

$$\mathcal{E}_n = X^{-1}(\mathcal{E}'_n), \qquad \mathcal{E} = \bigcap_{n=1}^{\infty} \mathcal{E}_n.$$

The σ-field $\mathcal{E}$ is the exchangeable σ-field. The tail field

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(X_{n+1}, X_{n+2}, \ldots)$$

satisfies $\mathcal{T} \subseteq \mathcal{E}$.

Theorem 3.55 (Finite-permutation averaging). Let (X_n) be exchangeable and let $\varphi : E^{\mathbb{N}} \to \mathbb{R}$ be integrable. For a permutation ρ, write $X^\rho = (X_{\rho(1)}, X_{\rho(2)}, \ldots)$. For $n \geq 1$, define

$$A_n\varphi = \frac{1}{n!} \sum_{\rho \in S_n} \varphi(X^\rho).$$

Then

$$A_n\varphi = \mathbb{E}[\varphi(X) \mid \mathcal{E}_n].$$

Proof. The average is $\mathcal{E}_n$-measurable. If $A \in \mathcal{E}_n$, then exchangeability and invariance of A give, for every $\rho \in S_n$,

$$\mathbb{E}[\varphi(X)\mathbf{1}_A] = \mathbb{E}[\varphi(X^\rho)\mathbf{1}_A].$$

Average over ρ to obtain the defining identity for conditional expectation. □

Theorem 3.56 (Reverse martingale convergence). Let $\mathcal{G}_1 \supseteq \mathcal{G}_2 \supseteq \cdots$ and $\mathcal{G}_\infty = \bigcap_n \mathcal{G}_n$. For $Z \in L^1$,

$$\mathbb{E}[Z \mid \mathcal{G}_n] \longrightarrow \mathbb{E}[Z \mid \mathcal{G}_\infty] \quad \text{almost surely and in } L^1.$$

More generally, if (Y_n) is adapted to the decreasing filtration and

$$\mathbb{E}[Y_n \mid \mathcal{G}_{n+1}] = Y_{n+1},$$

then (Y_n) is called a reverse martingale and has the displayed form with $Z = Y_1$.

Proof. The conditional expectations are uniformly integrable. To obtain almost sure convergence, fix N and reverse the finite sequence:

$$Z_k^{(N)} = \mathbb{E}[Z \mid \mathcal{G}_{N-k}], \qquad 0 \le k \le N-1.$$

With the increasing filtration $\mathcal{G}_{N-k}$, this is an ordinary martingale. Doob's upcrossing inequality applied to this reversed martingale and to its negative gives bounds, independent of N, on the expected numbers of upcrossings and downcrossings of every rational interval by the original sequence. Letting $N \to \infty$ and using the deterministic crossing criterion shows that the original sequence converges almost surely. Uniform integrability then gives L^1 convergence.
If (Y_n) is a reverse martingale, iteration of the tower property gives $Y_n = \mathbb{E}[Y_1 \mid \mathcal{G}_n]$. Finally, testing the L^1 limit against events in $\mathcal{G}_\infty$ identifies it as $\mathbb{E}[Z \mid \mathcal{G}_\infty]$. □

Theorem 3.57 (Exchangeable averages and the tail field). Let (X_n) be exchangeable. Let $k \ge 1$ and let $\varphi : E^k \to \mathbb{R}$ satisfy $\mathbb{E}|\varphi(X_1, \ldots, X_k)| < \infty$. For $n \ge k$, write

$$A_n\varphi = \frac{1}{(n)_k} \sum_{\substack{1 \le i_1, \ldots, i_k \le n \\ i_1, \ldots, i_k \text{ distinct}}} \varphi(X_{i_1}, \ldots, X_{i_k}),$$

where $(n)_k = n(n-1)\cdots(n-k+1)$. Then

$$A_n\varphi \longrightarrow \mathbb{E}[\varphi(X_1, \ldots, X_k) \mid \mathcal{E}] = \mathbb{E}[\varphi(X_1, \ldots, X_k) \mid \mathcal{T}]$$

almost surely and in L^1.

Proof. The ordered-tuple average is the same as the permutation average, so it is $\mathbb{E}[\varphi(X_1, \ldots, X_k) \mid \mathcal{E}_n]$. Since $\mathcal{E}_n \downarrow \mathcal{E}$, reverse martingale convergence gives the first limit.
Fix ℓ. Let $B_{n,\ell}$ be the same ordered-tuple average but using only indices in $\{\ell+1, \ldots, n\}$. It

is measurable with respect to $\sigma(X_{\ell+1}, X_{\ell+2}, \ldots)$. Exchangeability and a coupling of a uniform ordered k-tuple from $\{1, \ldots, n\}$ with one avoiding the first ℓ indices give

$$\|A_n\varphi - B_{n,\ell}\|_1 \le 2\mathbb{E}|\varphi(X_1, \ldots, X_k)| P(\text{the tuple uses one of } 1, \ldots, \ell) \longrightarrow 0.$$

Thus the L^1 limit L is measurable, in the L^1 sense, with respect to every $\sigma(X_{\ell+1}, X_{\ell+2}, \ldots)$; equivalently, $\mathbb{E}[L \mid \sigma(X_{\ell+1}, \ldots)] = L$. Applying reverse martingale convergence to L shows $L = \mathbb{E}[L \mid \mathcal{T}]$, so L has a tail measurable version. This proves the equality of the two conditional expectations. □

Theorem 3.58 (Exchangeable and tail fields agree modulo null sets). If (X_n) is exchangeable, then for every $A \in \mathcal{E}$ there is $B \in \mathcal{T}$ such that

$$P(A \triangle B) = 0.$$

Proof. Finite-coordinate bounded functions are dense in $L^1(\sigma(X_1, X_2, \ldots))$. Choose such functions φ_m with $\|\varphi_m - \mathbf{1}_A\|_1 \to 0$. The preceding theorem gives

$$\mathbb{E}[\varphi_m \mid \mathcal{E}] = \mathbb{E}[\varphi_m \mid \mathcal{T}].$$

Conditional expectation is an L^1 contraction, so letting $m \to \infty$ yields

$$\mathbf{1}_A = \mathbb{E}[\mathbf{1}_A \mid \mathcal{E}] = \mathbb{E}[\mathbf{1}_A \mid \mathcal{T}] \quad \text{almost surely.}$$

Choose a tail-measurable version Y of the last conditional expectation and let $B = \{Y > 1/2\}$. □

Corollary 3.59 (Hewitt–Savage zero–one law). If $X_1, X_2, \ldots$ are i.i.d., then every exchangeable event A satisfies

$$P(A) \in \{0, 1\}.$$

Proof. By the preceding theorem, A agrees almost surely with a tail event. Kolmogorov's zero–one law applies to the tail event. □